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CALCULUS 2

LECTURE NOTE

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1

Multivariable Functions

1.1 Multivariable Functions and Level Sets

¹ I do expect you to have learned about a single variable functions when you were in the high school. Recall that functions of single variable were just associations of a value of domain to the value of codomain:

¹ Lecture date: 2025-09-05

$$x \mapsto f(x)$$

where $x \in U \subseteq \mathbb{R}$. The idea of multivariable function is same.

Definition 1.1. Let $(x_1, \dots, x_n) \in V \subseteq \mathbb{R}^n$ and $n \geq 2$. We call f as a multivariable function if it gives an association between $(x_1, \dots, x_n)$ and $f(x_1, \dots, x_n)$, i.e.

$$(x_1, \dots, x_n) \mapsto f(x_1, \dots, x_n)$$

Note that this association should be well-defined.

Also, you must have drawn a graph of any single variable functions. But, do you know what the definition of graph is? To say that, we need to discuss about one more concept, the level set.

Recall how you have represented a sphere of radius r in $\mathbb{R}^3$. Maybe you think as:

$$x^2 + y^2 + z^2 = r^2$$

Yes, this is right. But, we can generalize this concept. Define $f(x, y, z) = x^2 + y^2 + z^2$. Then we get a function² $f : \mathbb{R}^3 \rightarrow \mathbb{R}$. Then one can view the sphere of radius

² From now on, "function" in this note denotes "multivariable function" unless otherwise specified.

r as a **solution set** of the equation

$$f(x, y, z) = r^2.$$

As you can see, the shape of this solution set can be different by taking another value. In general, we call this solution set as a level set.

Definition 1.2. Let $f : X \rightarrow Y$ be a function. Then for a constant $c \in Y$, the level set of f about c is defined as:

$$f^{-1}(c) := \{x \in X : f(x) = c\}.$$

Note that X and Y can be any spaces.

Then we can define a graph of a function.

Definition 1.3. Let $f : X \rightarrow Y$ be a function. Then the graph of f is:

$$\{(x, y) \in X \times Y : x \in f^{-1}(y)\}.$$

In particular, if $X \subseteq \mathbb{R}^n$ and $Y \subseteq \mathbb{R}$, then the graph of f is:

$$\{(x_1, \dots, x_n, y) \in \mathbb{R}^{n+1} : (x_1, \dots, x_n) \in f^{-1}(y)\}.$$

Equivalently, if we define $F : \mathbb{R}^{n+1} \rightarrow \mathbb{R}$ as $F(x_1, \dots, x_n, y) = f(x_1, \dots, x_n) - y$, then the graph of f is the level set $F^{-1}(0)$.

³ Note that, in the latter case, if you want to draw a graph of $f : \mathbb{R}^n \rightarrow \mathbb{R}$, you need one additional axis. That is, the maximum number of variables of a function which humans can draw a graph of it is 2, since we can only see up to 3-dimensional space. The reason why we cannot intuitively feel the sphere as a one part of “graph” is this; because we need 4 directions!

³ For example, imagine that you want to draw a graph of $y = x^3$. Then one may define $F(x, y) = x^3 - y$, and can observe that the usual graph you have known agrees with $F^{-1}(0)$.

1.2 Continuous Functions

We will just keep extending the concept which was defined in single variable case. Even though we can define the continuity in general space, we only consider the case where $f : \mathbb{R}^n \rightarrow \mathbb{R}$. ⁴

⁴ This is because we need to discuss the structure of space in general case, which is called **topology**. To avoid this, we only consider easier case.

Definition 1.4. Let $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$, and $P \in U$. We say that f is continuous at P if

$$\lim_{X \rightarrow P} f(X) = f(P).$$

We say f is continuous if it is continuous at every point in U .

Note that we need to consider all the directions of $X \rightarrow P$.⁵ Then by the definition, we can easily show the following theorem.

⁵ Therefore we usually use some inequalities.

Theorem 1.1. The followings hold.

1. Constant functions are continuous.
2. The prodiction, sum, difference, and quotient(when the denominator is non-zero) of continuous functions are continuous.
3. The composition of continuous functions are continuous.
4. All polynomials are continuous.

Finally, one can show the following theorem, which is non-trivial for now.⁶

⁶ I will omit the proof for now.

Theorem 1.2. Let $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ be a continuous function, where U is closed and bounded. Then f has the maximum, and the minimum.

⁷ ⁸ Note that some of discussions here are not that rigorous for better understanding.

⁷ Here, "closed" means the set contains all its boundary points, and "bounded" means the set is contained in some ball with finite radius.

⁸ Q) Is there a set simultaneously closed and open?

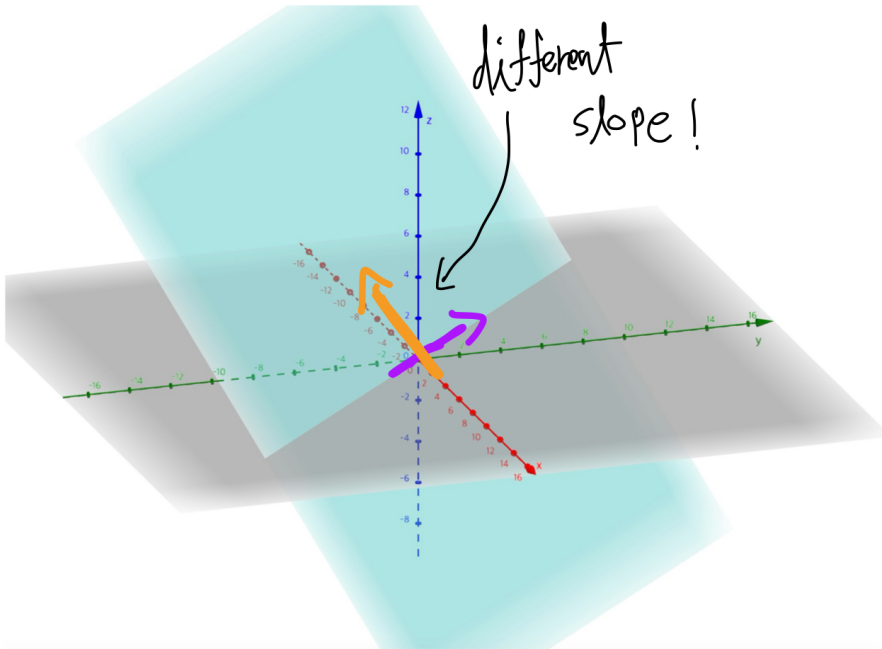
1.3 Slope of Functions of Degree One

As we treated a slope of a single variable function, we can discuss about a slope of a multivariable function too. But, it is slight different from the 1-dimensional case. Recall the definition of a slope(derivative) of a single variable function f at some point a :

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}.$$

As you can see, we added a "small amount" h , and divided by it again. This gave us the definition of a slope. But, in multivariable case, we need a direction of a

“small amount”, since the domain of f is not 1-dimensional. For example, consider the graph of a simple function $z = -x - y$.



You may clearly check that it has different slopes due to the direction you are looking at. Therefore, we need to say the **direction** when we want to say a direction.

Definition 1.5. Let $\mathbf{v}, \mathbf{x} \in \mathbb{R}^n$ be a unit vector. Then the slope(directional derivative) at $\mathbf{x}$ along $\mathbf{v}$ is:

$$D_{\mathbf{v}}f(\mathbf{x}) := \lim_{t \rightarrow 0} \frac{f(\mathbf{x} + t\mathbf{v}) - f(\mathbf{x})}{t}.$$

Even though there are very many multivariable functions, we specially call a function of the form

$$f(x_1, \dots, x_n) = a_1x_1 + a_2x_2 + \dots + a_nx_n + b = \mathbf{a} \cdot \mathbf{x} + b$$

as a function of degree one⁹. We only treat these functions in this section.

Theorem 1.3. The $\mathbf{v}$ -directional derivative of a function of degree one

$$z = \mathbf{a} \cdot \mathbf{x} + b = a_1x_1 + \dots + a_nx_n + b$$

⁹ Someone may one to argue that I didn't define the degree yet. You are right, but for now, just take it as a name.

equals to

$$\mathbf{a} \cdot \mathbf{v} = a_1 v_1 + \cdots + a_n v_n.$$

Proof. Direct from the definition. $\square$

1.4 Problems

Problem 10.1.6. Draw the level set of given f which contains P .

1. $f(x, y) = x + y, \quad P = (1, 1)$

2. $f(x, y) = x^3 + x^2 - y^2, \quad P = (0, 0)$

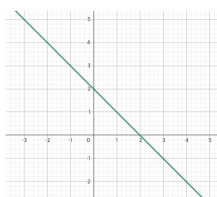
3. $f(x, y, z) = x^2 + y^2 - z^2, \quad P = (1, 0, 0)$

4. $f(x, y, z) = x^2 - y^2 - z^2, \quad P = (1, 0, 0)$

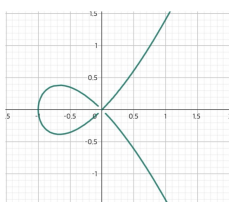
5. $f(x, y, z) = x^2 + 4y^2 + 9z^2, \quad P = \left(\frac{1}{\sqrt{3}}, \frac{1}{2\sqrt{3}}, \frac{1}{3\sqrt{3}}\right)$

Proof.

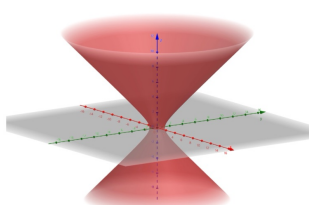
(1) $x + y = 2$



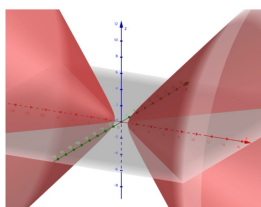
(2) $x^3 + x^2 - y^2 = 0$



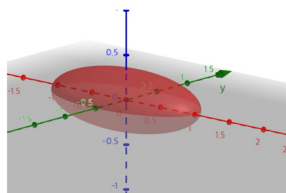
(3) $x^2 + y^2 - z^2 = 1$



(4) $x^2 - y^2 - z^2 = 1$



(5) $x^2 + 4y^2 + 9z^2 = 1$



$\square$

Problem 10.1.7. Let $f(x, y, z) = x + 2y + 3z + 4$, and $\mathbf{v} = \frac{1}{\sqrt{3}}(1, 1, 1)$. Compute the directional derivative of f along $\mathbf{v}$.¹⁰

¹⁰ Note that the point we take doesn't matter.

Proof. Observe that $f = (1, 2, 3) \cdot (x, y, z) + 4$. Then by the theorem 1.1, we get

$$(1, 2, 3) \cdot \mathbf{v} = \frac{6}{\sqrt{3}} = 2\sqrt{3}.$$

□

Problem 10.2.2. Determine whether the following functions are continuous.

$$1. \quad f(x, y) = \begin{cases} \frac{xy^2}{x^2 + y^4} & (x, y) \neq (0, 0), \\ 0 & (x, y) = (0, 0) \end{cases}$$

$$2. \quad f(x, y) = \begin{cases} \frac{x^2 - y^3}{x^2 + y^2} & (x, y) \neq (0, 0), \\ 0 & (x, y) = (0, 0) \end{cases}$$

$$3. \quad f(x, y) = \begin{cases} \frac{x^2y - xy^2}{x^2 + y^2} & (x, y) \neq (0, 0), \\ 0 & (x, y) = (0, 0) \end{cases}$$

Proof. By the theorem 1.2, suffice to check only at the origin.

1. If the domain is restricted to the parabola $x = y^2$, then for $y \neq 0$ the value of $f(x, y)$ is

$$f(y^2, y) = \frac{y^4}{2y^4} = \frac{1}{2}.$$

Then one can observe that $\lim_{y \rightarrow 0} f(y^2, y) \neq f(0, 0)$. Therefore, it is discontinuous.

2. If the domain is restricted to the line $y = 0$, then for $x \neq 0$ the value of $f(x, y)$ is

$$f(x, 0) = \frac{x^2 - 0}{x^2 + 0} = 1.$$

Then one can observe that $\lim_{x \rightarrow 0} f(x, 0) \neq f(0, 0)$. Therefore, it is discontinuous.

3. Since $x^2y - xy^2 = xy(x - y)$ and $|xy| \leq \frac{1}{2}(x^2 + y^2)$, it follows that

$$|f(x, y)| \leq \frac{1}{2}|x - y|.$$

Therefore, we get:

$$\lim_{(x,y) \rightarrow (0,0)} |f(x,y)| \leq \lim_{(x,y) \rightarrow (0,0)} \frac{1}{2} |x-y| = 0,$$

hence f is continuous.

□

Problem 10.2.3. Define

$$f(x,y) = \begin{cases} \frac{x^2 y}{x^4 + y^2} & (x,y) \neq (0,0) \\ 0 & o.w. \end{cases}$$

Show that f is continuous if it is restricted to any line, but not continuous if it is restricted to the parabola $y = x^2$.

Proof. Let $y = ax$. Then we get:

$$f(x, ax) = \frac{ax^3}{x^4 + a^2 x^2},$$

which implies

$$\lim_{x \rightarrow 0} f(x, ax) = \lim_{x \rightarrow 0} \frac{ax^3}{x^4 + a^2 x^2} = 0 = f(0,0).$$

Since a was arbitrary, we are done. Finally, put $y = x^2$. Then we get:

$$f(x, x^2) = \frac{x^4}{2x^4} = \frac{1}{2},$$

which implies

$$\lim_{x \rightarrow 0} f(x, x^2) = \frac{1}{2} \neq f(0,0).$$

Therefore, we are done.

□

Problem 10.2.4. Define

$$f(x, y) = \frac{ax + by}{x^2 + y^2}$$

for $(x, y) \neq (0, 0)$. What is the necessary and sufficient condition for f to be continuous at $(0, 0)$?

Proof. Again by the theorem 1.2, we suffice to check only at the origin. First suppose that f is continuous. Restrict the domain on $x = 0$ and $y = 0$, then one can observe that $a = b = 0$ is necessary.

Conversely, suppose that $a = b = 0$. Then $f(x, y) = 0$ for all $(x, y) \neq (0, 0)$, hence it is the necessary and sufficient condition which makes f continuous. $\square$

1.5 Differentiable Manifolds

Until now, the only spaces you treated are maybe $\mathbb{R}^n$'s. But in general, there are many spaces which are not identical with $\mathbb{R}^n$'s.¹¹ Among such spaces, one can consider some spaces which are locally similar to $\mathbb{R}^n$. One may ask "What does locally similar mean?". First, recall the definition of the open ball:

Definition 1.6. We call the following set in $\mathbb{R}^n$ as an open ball:

$$\mathbb{B}^n(P, r) := \{X \in \mathbb{R}^n : \|X - P\| < r\}$$

Using this, we can characterize the open sets in $\mathbb{R}^n$:

Definition 1.7. We call $U \subseteq \mathbb{R}^n$ as an open set if it is a union of open balls. Equivalently, U is open if for every $P \in U$, there exists $r > 0$ such that $\mathbb{B}^n(P, r) \subseteq U$.

The above is the standard definition of open sets in $\mathbb{R}^n$. We say "defining sets to be called open" as giving a **topology** to a given space. That is, one can change the definition of open sets, and that can define the very different space from the original space.

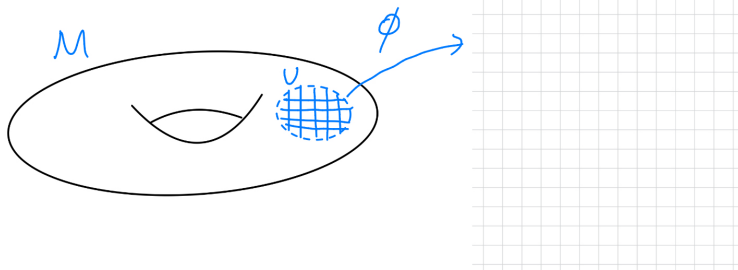
In short, one can understand that the **topology** contains the information of open sets of a given space.¹² This allows us to say "an open set" in an arbitrary space. Using this, we can define:

Definition 1.8. Let M be a topological space with a good topology. M is a **manifold** if for all $p \in M$, there exists an open set $U \ni p$ and a map $\phi : U \rightarrow \mathbb{R}^n$ where ϕ is bijective, continuous, and also its inverse ϕ^{-1} is continuous for some n . n is the dimension of the manifold, and we call M as an n -dimensional manifold.

Lecture date: 2025-09-12

¹¹ Here, just understand "not identical" as not looks similar. Note that this is very non-rigorous definition.

¹² For details, read little documents I uploaded, or [Munkres, Topology], and watch my additional video.



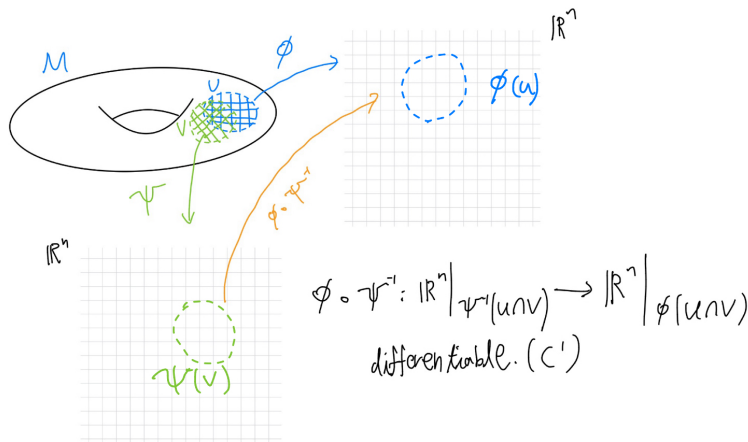
Example 1.5.1. Circles, spheres, and $\mathbb{R}^n$ itself, etc...

Now, one may have a very natural question after knowing what is the general definition of continuous: “Can we define differentiability in general spaces?”.

The answer is yes. Until the chapter 12, what we are going to do is studying the derivative of a function $f : M \rightarrow \mathbb{R}$ for some manifold M .

But, as you saw in the various examples, there are also many directions at a point in a manifold too. Then one may ask how can we define such “directions” in a manifold. To say that, we define the followings:

Definition 1.9. Let M be an n -dimensional manifold. We call M as a **differentiable manifold**(resp. **C^1 manifold**) if for all open sets $U, V \subseteq M$ where $U \cap V \neq \emptyset$ and for all homeomorphisms $\psi : V \rightarrow \mathbb{R}^n$, and $\phi : U \rightarrow \mathbb{R}^n$, the map $\phi \circ \psi^{-1} : \mathbb{R}^n|_{\psi^{-1}(U \cap V)} \rightarrow \mathbb{R}^n|_{\phi^{-1}(U \cap V)}$ is differentiable(resp. C^1).



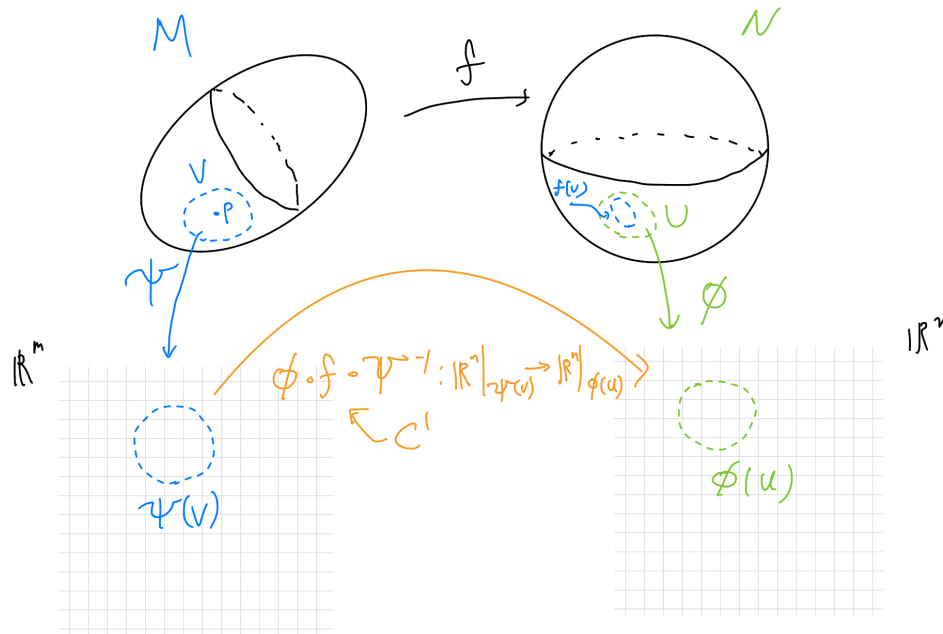
Roughly speaking, just think that a differentiable or C^1 manifold as a manifold with no issues with differentiability.^{13 14}

Definition 1.10. Let M be an m -dimensional C^1 manifold and N be an n -dimensional C^1 manifold. We call $f : M \rightarrow N$ as a C^1 at $p \in M$ if for some open $V \ni p$ and some open $U \subset N$ such that $f(V) \subseteq U$, and corresponding maps $\psi : V \rightarrow \mathbb{R}^m$, and $\phi : U \rightarrow \mathbb{R}^n$, $\phi \circ f \circ \psi^{-1} : \mathbb{R}^m|_{\psi(V)} \rightarrow \mathbb{R}^n|_{\phi(U)}$ is C^1 at $\psi(p)$. We call f is C^1 if it is C^1 at all points in M .

Recall a map $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is differentiable if its graph has tangent planes at each points and C^1 if all of its partial derivatives exist and are continuous.

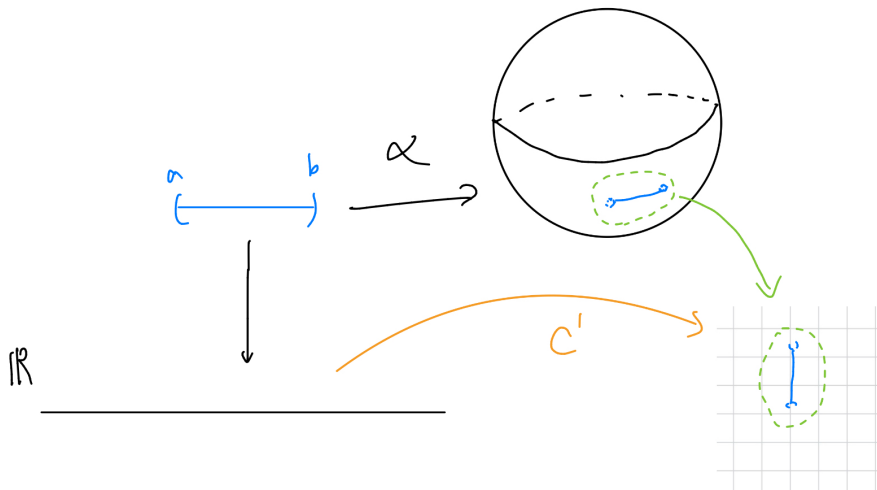
¹³ In fact, differentiable and C^1 manifolds are equivalent, moreover, they are all smooth manifolds which I will define far more days later.

¹⁴ By the sidenote above, I will just use C^1 manifolds only.



Now there is only one step left to define the “direction”.

Definition 1.11. Let M be an n -dimensional C^1 manifold, and $\alpha: (a, b) \rightarrow M$ be a C^1 map. We call such α as a **curve** in M .

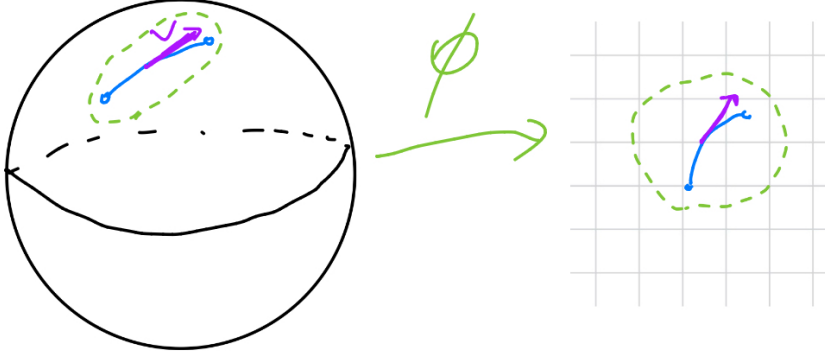


Using this, we can define the “direction” at a point in a manifold, a tangent vector.

Definition 1.12. Let M be an n -dimensional C^1 manifold. A **tangent vector** of M at $p \in M$ is $\alpha'(0)$ of a C^1 curve such that $\alpha : (-\epsilon, \epsilon) \rightarrow M$ with $\alpha(0) = p$, where ϵ is a sufficiently small positive.

Here, note that $\alpha'(0)$ is defined with some open set $p \in U$, and a corresponding map $\phi : U \rightarrow \mathbb{R}^n$.¹⁵

¹⁵ This $\alpha'(0)$ is independent to the choice of U and the corresponding map $\phi : U \rightarrow \mathbb{R}^n$.



As you can see in the figure above, for n -dimensional manifold, we have n -directions. Therefore, for each points, the vector space of tangent vectors is n -dimensional. For later, we will define the following:

Definition 1.13. The **tangent space** of M at $p \in M$ is the vector space defined by:

$$T_p M := \{\alpha'(0) : \alpha : (-\epsilon, \epsilon) \rightarrow M\}$$

If M is of n -dimension, then $T_p M$ is an n -dimensional vector space.

Now finally, we can define the directional derivative of a function $f : M \rightarrow \mathbb{R}$. Note that, the “direction” here should be one of tangent vectors of M (since f is only defined on M).

Note that $T_p M$ is a vector space, its basis would play very important role later.

Definition 1.14. Let M be an n -dimensional C^1 manifold, and $f : M \rightarrow \mathbb{R}$ be a C^1 function. Then the **directional derivative** of f at $p \in M$ along a tangent vector $\mathbf{v} \in T_p M$ is defined as follows: For some curve $\alpha : (-\epsilon, \epsilon) \rightarrow M$ such that $\alpha(0) = p$ and $\alpha'(0) = \mathbf{v}$, we define

$$D_{\mathbf{v}}f(p) := \left. \frac{d}{dt} \right|_0 (f \circ \alpha)(t) = (f \circ \alpha)'(0)$$

Theorem 1.4. Let M be an n -dimensional C^1 manifold, $f : M \rightarrow \mathbb{R}$ be C^1 , and $p \in M$. For $v, w \in T_p M$,

$$D_{v+w}f(p) = D_vf(p) + D_wf(p),$$

and for $c \in \mathbb{R}$,

$$D_{cv}f(p) = cD_vf(p).$$

That is, for fixed f and p , one may view $D_{(\cdot)}f(p) : T_p M \rightarrow \mathbb{R}$ as a linear map.

Proof sketch. Use the fact that $T_p M$ is a vector space. □

Now, we view the very special case: assume that $M = \mathbb{R}^n$. Clearly $\mathbb{R}^n$ is an n -dimensional C^1 manifold. Also, its tangent space at any $p \in \mathbb{R}^n$ is also $\mathbb{R}^n$. So, in this case, we may view $D_{(\cdot)}f(p) : \mathbb{R}^n \rightarrow \mathbb{R}$ as a linear map for fixed $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $p \in \mathbb{R}^n$.

Then what can we do? As we learned in the calculus 1 class, we may represent this $D_{(\cdot)}f(p)$ as a $1 \times n$ matrix. We define that matrix as:

Definition 1.15. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a C^1 function, and $p \in \mathbb{R}^n$. The **gradient** of f at p is an $1 \times n$ matrix representing $D_{(\cdot)}f(p) : \mathbb{R}^n \rightarrow \mathbb{R}$, that is:

$$D_{(v)}f(p) = Av = \text{grad } f(p)v$$

Since, in this case, $\text{grad } f(p)$ is $1 \times n$ matrix, we often call this as a **gradient vector**. Also, since the matrix multiplication between vectors are just dot product, we may write it as:

$$D_{(v)}f(p) = \text{grad } f(p) \cdot v$$

1.6 Partial Derivatives and Gradient Vectors

Now, within those definitions, we may clarify what the partial derivatives and gradient vectors are. Recall the definition of partial derivatives:

Definition 1.16. The k -th partial derivative of $f : \mathbb{R}^n \rightarrow \mathbb{R}$ at $p \in \mathbb{R}^n$ is $D_{e_k}f(p)$.

To calculate $D_{e_k}f(p)$, we need a path $\alpha : (\epsilon, \epsilon) \rightarrow \mathbb{R}^n$ such that $\alpha(0) = p$ and $\alpha'(0) = e_k$. Then clearly, one may take this path as:

$$\alpha(t) = (p_1, p_2, \dots, p_{k-1}, p_k + t, p_{k+1}, \dots, p_n).$$

Therefore, we get, just by the definition:

$$\begin{aligned} D_{e_k}f(p) &= \left. \frac{d}{dt} \right|_0 (f \circ \alpha)(t) \\ &= \left. \frac{d}{dt} \right|_0 f(p_1, p_2, \dots, p_{k-1}, p_k + t, p_{k+1}, \dots, p_n) \\ &= \lim_{t \rightarrow 0} \frac{f(p_1, p_2, \dots, p_{k-1}, p_k + t, p_{k+1}, \dots, p_n) - f(p)}{t} \\ &=: \frac{\partial f}{\partial x_k}(p). \end{aligned}$$

Now we may clarify what the gradient vector is. If we write $\text{grad } f(p) = (a_1, \dots, a_n)$, then one may notice that $\text{grad } f(p) \cdot e_k = a_k$. But then:

$$\begin{aligned} D_{e_k}f(p) &= \frac{\partial}{\partial x_k} f(p) \\ &= \text{grad } f(p) \cdot e_k = a_k \end{aligned}$$

Therefore, we get:

$$\text{grad } f(p) = (D_{e_1}f(p), D_{e_2}f(p), \dots, D_{e_n}f(p)) = \left(\frac{\partial f}{\partial x_1}(p), \frac{\partial f}{\partial x_2}(p), \dots, \frac{\partial f}{\partial x_n}(p) \right).$$

Here e_k is the k -th standard unit vector of $\mathbb{R}^n$.

1.7 Problems

Problem 10.3.2. At the point P , compute the directional derivative of f in the direction $\mathbf{v}$.

1. $f(x, y) = x^2 + y^3$, $P = (-1, 3)$, $\mathbf{v} = (1, 2)$

2. $f(x, y) = 2x^2 + xy - y^2$, $P = (3, -2)$, $\mathbf{v} = (1, -1)$

3. $f(x, y) = e^x \log y$, $P = (0, e)$, $\mathbf{v} = (2, 1)$

4. $f(x, y) = e^x \sin y$, $P = (0, \pi/4)$, $\mathbf{v} = (2, 1)$

5. $f(x, y) = \log \sqrt{x^2 + y^2}$, $P = (1, 1)$, $\mathbf{v} = (2, 1)$

Proof. Take a path $\alpha(t) = P + t\mathbf{v}$. By definition, compute $\lim_{t \rightarrow 0} \frac{f(P + t\mathbf{v}) - f(P)}{t}$.

1. $\lim_{t \rightarrow 0} \frac{(-1+t)^2 + (3+2t)^3 - 28}{t} = 52$

2. $\lim_{t \rightarrow 0} \frac{2(3+t)^2 + (3+t)(-2-t) - (-2-t)^2 - 8}{t} = 3$

3. $\lim_{t \rightarrow 0} \frac{e^{2t} \log(e+t) - 1}{t} = 2 + \frac{1}{e}$

4. $\lim_{t \rightarrow 0} \frac{e^{2t} \sin(\frac{\pi}{4} + t) - \frac{\sqrt{2}}{2}}{t} = \frac{3\sqrt{2}}{2}$

5. $\lim_{t \rightarrow 0} \frac{\log \sqrt{(1+2t)^2 + (1+t)^2} - \log \sqrt{2}}{t} = \frac{3}{2}$

□

Problem 10.3.7. Show that, for the function $f(x, y) = xy^2$,

$$\frac{\partial f}{\partial x}(1, 1) = D_1 f(1, 1) = 1.$$

Also, for the function $g(y, x) = yx^2$, explain that

$$\frac{\partial g}{\partial x}(1, 1) = D_2 g(1, 1) = 2.$$

Is $f = g$?

Proof.

$$\begin{aligned}\frac{\partial}{\partial x} f(x, y) &= y^2 \\ \frac{\partial}{\partial x} g(y, x) &= 2xy\end{aligned}$$

Now just plug in $(1, 1)$, then $D_1 f(1, 1) = 1$ and $D_2 g(1, 1) = 2$. Since it is just a matter of notation, $f \neq g$. □

Problem 10.4.8. Let the temperature distribution is given by

$$f(x, y) = x^2y + xy^2 \quad (^\circ\text{C}).$$

Let $p=(2,3)$. Answer the following:

1. In which direction should he move to cool down as quickly as possible? What is the instantaneous rate of change of temperature then?
2. In which direction does it get hottest the fastest? Also, what is the instantaneous rate of change of temperature in that direction?
3. In which directions is the instantaneous rate of change zero?

Proof. Observe that $\text{grad } f(2, 3) = (21, 16)$.

1. The opposite direction against the gradient vector, that is $(-21, -16)$. The instantaneous rate of change is $-\sqrt{21^2 + 16^2} = -\sqrt{697}$.
2. The same direction with the gradient vector, that is $(21, 16)$. The instantaneous rate of change is $\sqrt{21^2 + 16^2} = \sqrt{697}$.
3. The directions orthogonal to the gradient vector, that is $(16, -21)$ and $(-16, 21)$.

□

Problem 10.4.9. *let the differentiable function $f(x, y)$ satisfy at the origin that the $(1, 2)$ -directional derivative is 1 and the $(2, 3)$ -directional derivative is 2. What is the $(3, 5)$ -directional derivative of this function at the origin? Also find the gradient vector $\text{grad } f(O)$.*

Proof.

$$D_{(3,5)}f(O) = D_{(1,2)}f(O) + D_{(2,3)}f(O) = 3$$

$$D_{(1,0)}f(O) = -3D_{(1,2)}f(O) + 2D_{(2,3)}f(O) = 1$$

$$D_{(0,1)}f(O) = 2D_{(1,2)}f(O) - D_{(2,3)}f(O) = 0$$

$$\Rightarrow \text{grad } f(O) = (1, 0).$$

□

1.8 Chain Rule

Lecture date: 2025-09-19

By the definition of the gradient, we can easily show the following theorem:

Theorem 1.5. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable, and $\alpha : I \rightarrow \mathbb{R}^n$ be a differentiable curve. Then:

$$\frac{d}{dt}(f \circ \alpha)(t) = \text{grad } f(\alpha(t)) \cdot \alpha'(t).$$

Proof. This is direct by the definition of the gradient. Both terms are equal to $D_{\alpha'(t)}f(\alpha(t))$. □

Using the chain rule, we can show the following theorem:

Theorem 1.6. $\text{grad } f(P)$ is perpendicular to the level set of f which contains P .

Before proving this, we should define what is “perpendicular”.

Definition 1.17. We say a vector v is **perpendicular** to a set S at $P \in S$ if it is perpendicular to all the tangent vectors of S at P .

Proof of Theorem 1.6. Let $\alpha : (-\epsilon, \epsilon) \rightarrow S$. By the definition of tangent space, we suffice to show that for any such curves α , $\text{grad } f(P) \cdot \alpha'(0) = 0$. Since $\alpha(t) \in S$, $f(\alpha(t)) = f(P)$. Differentiate it, then by the chain rule:

$$0 = \frac{d}{dt}(f \circ \alpha)(t) \Big|_{t=0} = \text{grad } f(\alpha(0)) \cdot \alpha'(0) = \text{grad } f(P) \cdot \alpha'(0).$$

Therefore, we are done. ¹⁶ □

¹⁶ Make sure that the gradient vector is not zero. Consider the sphere.

1.9 Problems

Problem 10.5.6. Define

$$f(x, y) := \int_{xy}^{x^2+y^2} e^{-(t-1)^2} dt.$$

Compute $\frac{\partial f}{\partial x}(1, 0)$ and $\frac{\partial f}{\partial y}(0, 1)$.

Proof. Let $F(t) = \int e^{-(t-1)^2} dt$. Then $f(x, y) = F(x^2 + y^2) - F(xy)$. By the chain rule, we get:

$$\begin{aligned} \frac{\partial f}{\partial x}(x, y) &= e^{-(x^2+y^2-1)^2} \cdot 2x - e^{-(xy-1)^2} \cdot y \\ \frac{\partial f}{\partial y}(x, y) &= e^{-(x^2+y^2-1)^2} \cdot 2y - e^{-(xy-1)^2} \cdot x \end{aligned}$$

Therefore, $\frac{\partial f}{\partial x}(1, 0) = 2$ and $\frac{\partial f}{\partial y}(0, 1) = 2$. □

Problem 10.5.16. Let $g : (0, \infty) \rightarrow \mathbb{R}$ be differentiable. Define

$$f(X) := g(|X|).$$

Show that $\text{grad } f(X)$ is parallel to X .

Proof. By the chain rule, we get:

$$\begin{aligned}\text{grad } f(X) &= g'(|X|) \cdot \text{grad } |X| \\ &= g'(|X|) \cdot \frac{X}{|X|}\end{aligned}$$

Since $g'(|X|)$ is a scalar, we are done.

□ Consider the problem 10.3.5.

Problem 10.6.17. Show that the tangent line of the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

at (x_0, y_0) is

$$\frac{x_0 x}{a^2} + \frac{y_0 y}{b^2} = 1.$$

Proof. We may consider it as a zero-level set of

$$f(x, y) = \frac{x^2}{a^2} + \frac{y^2}{b^2} - 1$$

By the theorem 1.6, the tangent line is perpendicular to $\text{grad } f(x_0, y_0) = \left(\frac{2x_0}{a^2}, \frac{2y_0}{b^2}\right)$.

Therefore,

$$\frac{x_0}{a^2}(x - x_0) + \frac{y_0}{b^2}(y - y_0) = 0$$

is the tangent line. □

Problem 10.6.3. 3. Find the equation of the tangent plane at the point P to the surface given by the following equation.

1. $8x^2 + y^2 + 8z^2 = 16$, $P = (1, 2, 1/\sqrt{2})$

2. $x^2 + y^2 - z^2 = 4$, $P = (2, 1, 1)$

3. $x^2 + \frac{1}{4}y^2 - \frac{1}{9}z^2 = 1$, $P = (1, 2, 3)$

4. $\sin xy + \sin yz + \sin zx = 1$, $P = (1, \frac{\pi}{2}, 0)$

5. $z - e^x \sin y = 0$, $P = (\log 3, \frac{3\pi}{2}, -3)$

6. $2y - z^3 - 3xz = 0$, $P = (1, 7, 2)$

7. $x^3 - 2y^2 + z^2 = 0$, $P = (1, 1, 1)$

Proof. Calculate the gradient at the given point.

	$\nabla f(x, y, z)$	Equation of the plane
(1)	$(16x, 2y, 16z)$	$16(x - 1) + 4(y - 2) + 8\sqrt{2}\left(z - \frac{1}{\sqrt{2}}\right) = 0$
(2)	$(2x, 2y, -2z)$	$4(x - 2) + 2(y - 1) - 2(z - 1) = 0$
(3)	$\left(2x, \frac{1}{2}y, -\frac{2}{9}z\right)$	$2(x - 1) + (y - 2) - \frac{2}{3}(z - 3) = 0$
(4)	$(y \cos xy + z \cos zx, x \cos xy + z \cos yz, y \cos yz + x \cos zx)$	$\left(\frac{\pi}{2} + 1\right)(z - 0) = 0$
(5)	$(-e^x \sin y, -e^x \cos y, 1)$	$3(x - \log 3) + (z + 3) = 0$
(6)	$(-3z, 2, -3z^2 - 3x)$	$-6(x - 1) + 2(y - 7) - 15(z - 2) = 0$
(7)	$(3x^2, -4y, 2z)$	$3(x - 1) - 4(y - 1) + 2(z - 1) = 0$

□

Problem 10.6.4. Calculate the tangent plane of the surfaces

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = -1$$

at the point (x_0, y_0, z_0) .

Proof. By the gradient vectors, we get:

$$\frac{x_0 x}{a^2} + \frac{y_0 y}{b^2} - \frac{z_0 z}{c^2} = 1$$

$$\frac{x_0 x}{a^2} + \frac{y_0 y}{b^2} - \frac{z_0 z}{c^2} = -1$$

□

Higher Order Derivatives

2.1 Leibniz Rule

As we did in the high school, there is a good relationship between the integration and the differentiation. But there are very strong tool which we didn't learn yet: the **Leibniz Rule**. This the problem 10.5.7,

Theorem 2.1. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a continuous function which is differentiable with respect to x . Also suppose that the partial derivative $\frac{\partial}{\partial x}f(x, y)$ is continuous for both variables x and y . Additionally, let $a(x)$ and $b(x)$ be C^1 functions. Then:

$$\frac{d}{dx} \int_{a(x)}^{b(x)} f(x, y) dy = f(x, b(x))b'(x) - f(x, a(x))a'(x) + \int_{a(x)}^{b(x)} \frac{\partial}{\partial x}f(x, y) dy.$$

Proof. Define

$$G(u, v, w) := \int_u^v g(w, y) dy$$

Now we apply the same idea with the problem 10.5.7. Observe that:

$$\begin{aligned} D_1 G(u, v, w) &= \frac{\partial}{\partial u} \int_u^v g(w, y) dy \\ &= -g(w, u) \\ D_2 G(u, v, w) &= \frac{\partial}{\partial v} \int_u^v g(w, y) dy \\ &= g(w, v) \\ D_3 G(u, v, w) &= \frac{\partial}{\partial w} \int_u^v g(w, y) dy \\ &= \int_u^v \frac{\partial}{\partial w} g(w, y) dy \end{aligned}$$

Therefore, we get:

$$\begin{aligned} \frac{d}{dx} \int_{a(x)}^{b(x)} g(x, y) dy &= \frac{d}{dx} G(a(x), b(x), x) \\ &= g(x, b(x))b'(x) - g(x, a(x))a'(x) + \int_{a(x)}^{b(x)} \frac{\partial}{\partial x} g(x, y) dy \end{aligned}$$

□

We can apply this theorem: the Feynmann technique.¹

¹ Reference: LEO GOLDBAKHER,
DIFFERENTIATION UNDER THE
INTEGRAL SIGN

2.2 Second Order Derivatives

We studied some properties of the gradient. But one may have observed that we only did something with first order differentiation. So what we are going to do now is to study higher order derivatives. From now on, we will write the vector space of functions which are differentiable k times as C^k . The most important thing in this section is followings:

Theorem 2.2. *Derivatives are independent of the order of differentiation. That is, for a C^k function f ,*

$$D_{i_1} D_{i_2} \cdots D_{i_n} f = D_{j_1} D_{j_2} \cdots D_{j_n} f.$$

Proof. Omit. Use the Leibniz rule. □

The following is a hint for the next lecture: recall that we observed that $D_v f(x) = \text{grad } f(x) \cdot v$. Therefore, we may write:

$$D_v = \sum_{i=1}^n v_i D_i = \sum_{i=1}^n v_i \frac{\partial}{\partial x_i}.$$

That is, we may use $\frac{\partial}{\partial x_i}$ as a coordinate system of D_v space. Intuitively, a general vector corresponds to a general arbitrary directional D_v , and the standard directions correspond to $D_i = \frac{\partial}{\partial x_i}$.

2.3 Problems

Problem 11.1.1. For $x > 0$, compute:

$$\frac{d}{dx} \int_{x^2}^{x^3} \frac{1}{y} e^{xy^2} dy$$

Proof. Just apply the Leibniz rule.

$$\begin{aligned} \frac{1}{x^3} e^{x \cdot x^3} 3x^2 - \frac{1}{x^2} e^{x \cdot x^2} 2x + \int_{x^2}^{x^3} y e^{xy^2} dy &= \frac{3e^{x^4} - 2e^{x^3}}{x} + \left[\frac{1}{2} e^{xy^2} \right]_{x^2}^{x^3} \\ &= \frac{3e^{x^4} - 2e^{x^3}}{x} + \frac{1}{2} (e^{x^7} - e^{x^5}) \end{aligned}$$

□

Problem 11.2.13. Suppose that $f \in C^k$. Show that for any $v = \sum v_i e_i$,

$$D_v^k f = \sum_{i_1, \dots, i_k=1}^n v_{i_1} \cdots v_{i_k} D_{i_1} \cdots D_{i_k} f$$

Proof. If $k = 1$, we already saw it. Now we use induction. Suppose that this holds for $k - 1$. Then:

$$\begin{aligned} D_v^k f &= D_v(D_v^{k-1} f) \\ &= D_v \left(\sum_{i_1, \dots, i_{k-1}=1}^n v_{i_1} \cdots v_{i_{k-1}} D_{i_1} \cdots D_{i_{k-1}} f \right) \\ &= \sum_{j=1}^n v_j D_j \sum_{i_1, \dots, i_{k-1}=1}^n v_{i_1} \cdots v_{i_{k-1}} D_{i_1} \cdots D_{i_{k-1}} f \\ &= \sum_{i_1, \dots, i_k=1}^n v_{i_1} \cdots v_{i_k} D_{i_1} \cdots D_{i_k} f \end{aligned}$$

□

2.4 Taylor Expansion

We learned the following in the calculus lecture class;

Theorem 2.3. *Let $U \subseteq \mathbb{R}^n$ such that $P \in U$. If $f \in C^k(U)$, then there exist a unique k -th degree polynomial $T_k(\mathbf{v})$ and a function $R_k(\mathbf{v}) \in o(|\mathbf{v}|^k)$ such that*

$$f(P + \mathbf{v}) = T_k(\mathbf{v}) + R_k(\mathbf{v}).$$

This allows us to whether the given polynomial is indeed k -th degree approximation polynomial by considering $o(\sqrt{x_1^2 + \cdots + x_n^2}^k)$.

2.5 Problems

Problem 11.3.1. Compute the second order Taylor expansion.

1. $\sin(xy)$
2. $\sin x \cos y$
3. $\sin(x^2 + y^2)$
4. $\sin(e^y + x^2 - 2)$
5. $\log(1 + xy)$
6. $x + xy + 2y^2 + x^2$
7. e^{x+y}
8. $e^x - e^y$
9. $\frac{1}{1 - x^2 - y^2}$
10. $e^x \tan y$
11. $(1 + x)^y$
12. $\sin e^{xy}$

Proof.

1. One may consider the following table:

$f(x, y)$	$\frac{\partial}{\partial x}$	$\frac{\partial}{\partial y}$	$\frac{\partial^2}{\partial x^2}$	$\frac{\partial}{\partial x} \frac{\partial}{\partial y}$	$\frac{\partial^2}{\partial y^2}$	$T_2 f(x, y)$
$\sin(xy)$	$y \cos(xy)$	$x \cos(xy)$	$-y^2 \sin(xy)$	$\cos(xy) - xy \sin(xy)$	$-x^2 \sin(xy)$	xy

Alternatively, one may show the following lemma.

Lemma 11.3.1. Let $p(X)$ be a polynomial, and let $\mathfrak{R}p(X)$ be sum of terms of $p(X)$ whose degree are less or equal to k . Also, let f, g be C^k functions such that $g(a) = b$. Write the Taylor expansions of degree k at $X = b$ and $X = a$ of them as:

$$f(X) = T_k f(X) + R_k f(X), \quad g(X) = T_k g(X) + R_k g(X)$$

Then the Taylor expansion of degree k of $f \circ g$ at $X = a$ is:

$$T_k(f \circ g) = \mathfrak{R}T_k f(T_k g)$$

Proof. Let $f \circ g = T_k(f \circ g) + R_k f \circ g$. By the Taylor expansion of f and g , we have:

$$\begin{aligned} f \circ g(X) &= T_k f(g(X)) + R_k f(g(X)) = T_k(T_k g + R_k g) + R_k f(g(X)) \\ &= T_k f(T_k g) + [T_k(T_k g + R_k g) - T_k f(T_k g)] + R_k f(g(X)) \end{aligned}$$

$$= \underbrace{\Re T_k f(T_k g)}_{\text{term 1}} + \underbrace{T_k f(T_k g) - \Re T_k f(T_k g)}_{\text{term 2}} + \underbrace{[T_k(T_k g + R_k g) - T_k f(T_k g)]}_{\text{term 3}} + \underbrace{R_k f(g(X))}_{\text{term 4}}$$

- (a) Term 1 is clearly a polynomial under degree k with terms $(X - a)$.
- (b) Term 2 is a polynomial whose degree is over k , i.e. $o(|X - a|^k)$.
- (c) By the MVT we have

$$T_k(T_k g + R_k g) - T_k f(T_k g) \leq \sup_{c \in B_{R_k g(X)}(T_k g(X))} |T'_k f(c)| \cdot |R_k g(X)|$$

Since $R_k g(X) = o(|X - a|^k)$, we are done.

- (d) By the definition of $R_k f$, $R_k f(g(X)) = o(|g(X) - b|^k)$. Observe that

$$g(X) - b = g(X) - g(a) = g'(a)(X - a) + o(|X - a|).$$

Therefore $R_k f(g(X)) = o(|g(X) - b|^k) = o(|X - a|^k)$. The uniqueness of the taylor expansion concludes the proof.

□

Now by the lemma, we may just composite: recall that $\sin x = x + o(x^2)$. Therefore, $\sin(xy) = xy + o(\sqrt{x^2 + y^2}^2)$. From now on, I will just write only one solution.

2. In the same way of the proof of lemma, one can show that the taylor approximation of the product of functions is just the product of taylor approximations of each function. Multiply the taylor approximation: $\sin x = x + o(x^2)$, $\cos y = 1 - \frac{1}{2}y^2 + o(y^2)$. We get

$$T_2 \sin x \cos y = x$$

3. Use the lemma: we get $T_2 \sin(x^2 + y^2) = x^2 + y^2$.
4. Note that $e^y + x^2 - 2|_{(0,0)} \neq 0$. Therefore we cannot apply the lemma in this case. Just differentiate, we get:

$f(x, y)$	$\frac{\partial}{\partial x}$	$\frac{\partial}{\partial y}$	$\frac{\partial^2}{\partial x^2}$
$\sin(e^y + x^2 - 2)$	$2x \cos(e^y + x^2 - 2)$	$e^y \cos(e^y + x^2 - 2)$	$2 \cos(e^y + x^2 - 2) - 4x^2 \sin(e^y + x^2 - 2)$
$\frac{\partial}{\partial x} \frac{\partial}{\partial y}$	$\frac{\partial^2}{\partial y^2}$		$T_2 f(x, y)$
$-2xe^y \sin(e^y + x^2 - 2)$	$e^y \cos(e^y + x^2 - 2) - e^{2y} \sin(e^y + x^2 - 2)$		$-\sin 1 + y \cos 1 + x^2 \cos 1 + \frac{1}{2}y^2(\cos 1 + \sin 1)$

5. $\log(1+x) = x + o(x^2)$. Therefore $T_2(\log(1+xy)) = xy$.

6. Itself.

7. Composition of e^x and $x+y$. Applying the lemma, we get:

$$T_2 e^{x+y} = 1 + (x+y) + \frac{1}{2}(x+y)^2$$

8. Subtract the Taylor approximation. We get $T_2 e^x - e^y = (x-y) + \frac{1}{2}(x^2 - y^2)$

9. Apply the lemma:

$$T_2 \frac{1}{1-x^2-y^2} = 1 + x^2 + y^2$$

10. Multiply. $T_2 e^x \tan y = y + xy$

11. Let $f(x, y) = (1+x)^y$.

$$\begin{aligned} \frac{\partial f}{\partial x} &= y(1+x)^{y-1}, \\ \frac{\partial f}{\partial y} &= (1+x)^y \log(1+x), \\ \frac{\partial^2 f}{\partial x^2} &= y(y-1)(1+x)^{y-2}, \\ \frac{\partial^2 f}{\partial x \partial y} &= (1+x)^{y-1} + y(1+x)^{y-1} \log(1+x), \\ \frac{\partial^2 f}{\partial y^2} &= (1+x)^y (\log(1+x))^2. \end{aligned}$$

Therefore $T_2 f = 1 + xy$.

12. Let $f = \sin e^{xy}$. Then

$$\begin{aligned} \frac{\partial f}{\partial x} &= ye^{xy} \cos e^{xy}, \\ \frac{\partial f}{\partial y} &= xe^{xy} \cos e^{xy}, \\ \frac{\partial^2 f}{\partial x^2} &= y^2 e^{xy} \cos e^{xy} - y^2 e^{2xy} \sin e^{xy}, \\ \frac{\partial^2 f}{\partial x \partial y} &= e^{xy} \cos e^{xy} + xye^{xy} \cos e^{xy} - xye^{2xy} \sin e^{xy}, \\ \frac{\partial^2 f}{\partial y^2} &= x^2 e^{xy} \cos e^{xy} - x^2 e^{2xy} \sin e^{xy}. \end{aligned}$$

Therefore $T_2 f = \sin 1 + xy \cos 1$.

□

Problem 11.3.12. Let $P \in \mathbb{R}^n$. Take a neighborhood U of P , and let $f, g \in C^k(U)$. If $f(P+v) - g(P+v) = o(|v|^k)$, show that:

$$T_k f(P, v) = T_k g(P, v)$$

Proof. Write $f(P+v) = T_k f(P, v) + R_k f(P, v)$, and $g(P+v) = T_k g(P, v) + R_k g(P, v)$. Subtract them:

$$f(P+v) - g(P+v) = \underbrace{[T_k f(P, v) - T_k g(P, v)]}_{\text{polynomial under degree } k} + \underbrace{[R_k f(P, v) - R_k g(P, v)]}_{\in o(|v|^k)}$$

By the assumption that $f(P+v) - g(P+v) = o(|v|^k)$, we get $T_k f(P, v) - T_k g(P, v) = 0$.

□

Problem 11.4.1. Find the critical points of the following functions.

- | | |
|---------------------------------|---------------------------|
| (1) $x^2 + 4xy - y^2 - ax - by$ | (2) $x - y \sin x$ |
| (3) x^2y^2 | (4) $x^4 + y^2$ |
| (5) $e^{x^2+y^2+z^2}$ | (6) $e^{-xy/(x^2+y^2+1)}$ |

Proof. For each problem, let the given function be denoted by f .

- (1) $\text{grad } f = (2x + 4y - a, 4x - 2y - b)$ so the critical point is $\left(\frac{a + 2b}{10}, \frac{2a - b}{10}\right)$.
- (2) $\text{grad } f = (1 - y \cos x, -\sin x)$ so the critical points are $(n\pi, (-1)^n)$ (where $n \in \mathbb{Z}$).
- (3) $\text{grad } f = (2xy^2, 2x^2y)$ so the set of critical points is the x - and y -axes.
- (4) $\text{grad } f = (4x^3, 2y)$ so the critical point is $(0, 0)$.
- (5) $\text{grad } f = 2e^{x^2+y^2+z^2}(x, y, z)$ so the critical point is $(0, 0, 0)$.
- (6) $\text{grad } f = \frac{e^{-xy/(x^2+y^2+1)}}{(x^2 + y^2 + 1)^2} (y(x^2 - y^2 - 1), x(y^2 - x^2 - 1))$, so the critical point is $(0, 0)$.

□

Problem 11.4.3. Let $f(x, y) = 3x^4 - 4x^2y + y^2 = (3x^2 - y)(x^2 - y)$.

1. Show that the origin is a critical point of f .
2. For any unit vector $v = (h, k)$, show that the origin is a local minimum of $g(t) = f(tv)$.
3. Show that the origin is not a local minimum of f .
4. Draw $f = 0$.
5. Draw $f > 0$ and $f < 0$.
6. How can (2) and (3) are compatible?

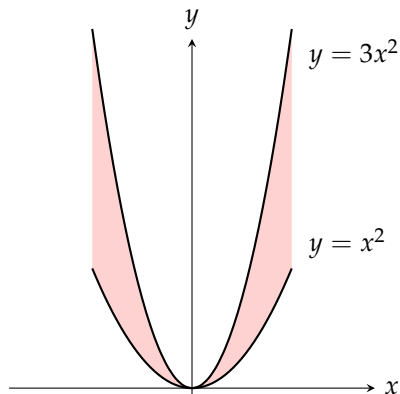
Proof.

1. $\text{grad } f = (12x^3 - 8xy, 4x^2 + 2y)$, hence $\text{grad } f(0, 0) = (0, 0)$.
2. Note that $h^2 + k^2 = 1$, in particular they cannot simultaneously 0.

$$\begin{aligned} g(t) &= 3h^4t^4 - 4h^2kt^3 + k^2t^2 \\ g'(t) &= 12h^4t^3 - 12h^2kt^2 + 2k^2t \\ g''(t) &= 36h^4t^2 - 24h^2kt + 2k^2 \end{aligned}$$

Therefore $g'(0) = 0$ and $g''(0) = 2k^2$. If $k \neq 0$, we are done. If $k = 0$, then $g(t) = 3t^4$, so the origin is a local minimum.

3. Consider the path $(t, 2t^2)$. $f(t, 2t^2) = -t^4$, hence the origin is not a local minimum.
4. Draw $y = x^2$ and $y = 3x^2$.
5. Red area is negative, and the remaining is positive.



6. Trivial.

□

Problem 11.5.1. *Define*

$$f(x, y) := x^4 + 2x^2y + 3xy^2 + 6xy + 3x$$

Calculate the maximal and minimal point of f if exist.

Proof.

$$f(x, 0) = x^4 + 3x$$

$$f(-1, y) = -3y^2 - 4y - 2$$

Therefore both of them do not exist.

□

Problem 11.5.4. Find the local maxima, local minima, and saddle points of the following functions.

- (1) $\frac{1}{x} + xy - \frac{8}{y}$, (2) $x^2 - y^2 - 2xy - 4x$,
 (3) $x^2 + 9 + 6x \cos y$, (4) $\log(1 + x^2 + y^2)$,
 (5) $4 \log x + e^y - x - y$, (6) $x^4 + y^4 - (x + y)^2$,
 (7) $x^2 + 2xy + 2y^2 + 4x$, (8) $x^3 - y^3 + 3x^2 + 3y^2 - 9x$,
 (9) $x^2 - xy + y^4$, (10) $\frac{x^3 - 3x}{1 + y^2}$,
 (11) $(x^2 - y^2)e^{-(x^2 + y^2)/2}$

Also answer for the following functions:

(12) $\sin x + \sin y + \sin(x + y)$, (13) $(x + y)^3 + (x - y)^3 - 12(x + y)$.

Proof. 1. Let $f(x, y) = \frac{1}{x} + xy - \frac{8}{y}$.

$$\text{grad } f = \left(-\frac{1}{x^2} + y, x + \frac{8}{y^2} \right)$$

So the critical point is $(-\frac{1}{2}, 4)$. The Hessian is

$$f''(x, y) = \begin{pmatrix} \frac{2}{x^3} & 1 \\ 1 & -\frac{16}{y^3} \end{pmatrix}, \quad f''(-\frac{1}{2}, 4) = \begin{pmatrix} -16 & 1 \\ 1 & -\frac{1}{4} \end{pmatrix}.$$

This is negative definite; hence $(-\frac{1}{2}, 4)$ is a local maximum.

2. Let $f(x, y) = x^2 - y^2 - 2xy - 4x$.

$$\text{grad } f = (2x - 2y - 4, -2y - 2x), \quad \text{critical point } (1, -1).$$

$$f''(x, y) = \begin{pmatrix} 2 & -2 \\ -2 & -2 \end{pmatrix}, \quad \det f''(1, -1) < 0.$$

Therefore $(1, -1)$ is a saddle point.

3. Let $f(x, y) = x^2 + 9 + 6x \cos y$.

$$\text{grad } f = (2x + 6 \cos y, -6x \sin y).$$

Critical points:

$$A_n = (3(-1)^{n+1}, n\pi), \quad B_n = \left(0, \frac{\pi}{2} + n\pi\right) \quad (n \in \mathbb{Z}).$$

Hessian and determinant:

$$f''(x, y) = \begin{pmatrix} 2 & -6 \sin y \\ -6 \sin y & -6x \cos y \end{pmatrix}, \quad \det f''(x, y) = -12x \cos y - 36 \sin^2 y.$$

We have

$$\det f''(A_n) = 36, \quad \det f''(B_n) = -36,$$

so A_n is a local minimum and B_n is a saddle point.

4. Let $f(x, y) = \log(1 + x^2 + y^2)$.

$$\text{grad } f = \left(\frac{2x}{1 + x^2 + y^2}, \frac{2y}{1 + x^2 + y^2} \right), \quad \text{critical point } (0, 0).$$

$$f''(x, y) = \frac{1}{(1 + x^2 + y^2)^2} \begin{pmatrix} 2 - 2x^2 + 2y^2 & -4xy \\ -4xy & 2 + 2x^2 - 2y^2 \end{pmatrix}.$$

Since $f''(0, 0)$ is positive definite, $(0, 0)$ is a local minimum.

5. Let $f(x, y) = 4 \log x + e^y - x - y$.

$$\text{grad } f = \left(\frac{4}{x} - 1, e^y - 1 \right), \quad \text{critical point } (4, 0).$$

$$f''(x, y) = \begin{pmatrix} -\frac{4}{x^2} & 0 \\ 0 & e^y \end{pmatrix}, \quad \det f''(4, 0) = -\frac{1}{4} < 0.$$

Hence $(4, 0)$ is a saddle point.

6. Let $f(x, y) = x^4 + y^4 - (x + y)^2$.

$$\text{grad } f = (4x^3 - 2(x + y), 4y^3 - 2(x + y)),$$

so the critical points are $(-1, -1)$, $(0, 0)$, $(1, 1)$.

$$f''(x, y) = \begin{pmatrix} 12x^2 - 2 & -2 \\ -2 & 12y^2 - 2 \end{pmatrix}.$$

Thus $(-1, -1)$ and $(1, 1)$ are local minima. For $(0, 0)$ the Hessian test is incon-

clusive. Let $F(t) = f(t, t) = 2t^4 - 4t^2$ (max at $t = 0$) and $G(t) = f(t, -t) = 2t^4$ (min at $t = 0$); hence the origin is a saddle point.

7. Let $f(x, y) = x^2 + 2xy + 2y^2 + 4x$.

$$\text{grad } f = (2x + 2y + 4, 2x + 4y), \quad \text{critical point } (-4, 2).$$

$$f''(x, y) = \begin{pmatrix} 2 & 2 \\ 2 & 4 \end{pmatrix},$$

which is positive definite; hence $(-4, 2)$ is a local minimum.

8. Let $f(x, y) = x^3 - y^3 + 3x^2 + 3y^2 - 9x$.

$$\text{grad } f = (3x^2 + 6x - 9, -3y^2 + 6y),$$

so the critical points are $(-3, 0)$, $(-3, 2)$, $(1, 0)$, $(1, 2)$.

$$f''(x, y) = \begin{pmatrix} 6x + 6 & 0 \\ 0 & -6y + 6 \end{pmatrix}, \quad \det f''(x, y) = 36(x + 1)(-y + 1).$$

Therefore $(-3, 0)$ and $(1, 2)$ are saddle points, $(-3, 2)$ is a local maximum, and $(1, 0)$ is a local minimum.

9. Since $\text{grad } f = (2x - y, -x + 4y^3)$, the critical points are $(0, 0)$ and $\pm\left(\frac{1}{4\sqrt{2}}, \frac{1}{2\sqrt{2}}\right)$.

The Hessian is

$$f''(x, y) = \begin{pmatrix} 2 & -1 \\ -1 & 12y^2 \end{pmatrix}.$$

Hence $\det f''(O) = -1$, so the origin is a saddle point, and

$$\det f''\left(\pm\left(\frac{1}{4\sqrt{2}}, \frac{1}{2\sqrt{2}}\right)\right) = 2,$$

so $\pm\left(\frac{1}{4\sqrt{2}}, \frac{1}{2\sqrt{2}}\right)$ are local minima.

10. Since

$$\text{grad } f = \left(\frac{3x^2 - 3}{1 + y^2}, -\frac{2y(x^3 - 3x)}{(1 + y^2)^2}\right),$$

the critical points are $\pm(1, 0)$. The Hessian is

$$f''(x, y) = \begin{pmatrix} \frac{6x}{1+y^2} & -\frac{2y(3x^2-3)}{(1+y^2)^2} \\ \frac{2y(3x^2-3)}{(1+y^2)^2} & \frac{(x^3-3x)(6y^2-2)}{(1+y^2)^3} \end{pmatrix},$$

and thus

$$f''(\pm(1, 0)) = \pm \begin{pmatrix} 6 & 0 \\ 0 & 4 \end{pmatrix}.$$

Therefore $(1, 0)$ is a local minimum, while $(-1, 0)$ is a local maximum.

11. Since

$$\text{grad } f = e^{-(x^2+y^2)/2} (x(y^2 - x^2 + 2), y(y^2 - x^2 - 2)),$$

the critical points are $(0, 0)$, $\pm(\sqrt{2}, 0)$, $\pm(0, \sqrt{2})$. The Hessian is

$$f''(x, y) = e^{-(x^2+y^2)/2} \begin{pmatrix} x^4 - x^2y^2 - 5x^2 + y^2 + 2 & xy(x^2 - y^2) \\ xy(x^2 - y^2) & -y^4 + x^2y^2 - x^2 + 5y^2 - 2 \end{pmatrix}.$$

Substituting each case shows that $(0, 0)$ is a saddle point, $\pm(\sqrt{2}, 0)$ are local maxima, and $\pm(0, \sqrt{2})$ are local minima.

12. Since

$$\text{grad } f = (\cos x + \cos(x + y), \cos y + \cos(x + y)),$$

at a critical point we must have $\cos x = \cos y$. Hence $x - y$ is a multiple of 2π , or $x + y$ is a multiple of 2π . Assume first that $x - y \equiv 0 \pmod{2\pi}$. Then from $\cos x + \cos(x + y) = \cos x + \cos 2x = 0$ we obtain $\cos x = -1$ or $\cos x = \frac{1}{2}$.

Therefore

$$(x, y) = (\pi + 2m\pi, \pi + 2n\pi), \quad (2p\pi + \frac{\pi}{3}, 2q\pi + \frac{\pi}{3}), \quad (2p\pi - \frac{\pi}{3}, 2q\pi - \frac{\pi}{3})$$

(with $m, n, p, q \in \mathbb{Z}$). When $x + y$ is a multiple of 2π , these are included in the above list, so the above exhaust all critical points.

(a) For $(x, y) = (2p\pi + \frac{\pi}{3}, 2q\pi + \frac{\pi}{3})$, the Hessian is

$$\frac{1}{2} \begin{pmatrix} -2\sqrt{3} & -\sqrt{3} \\ -\sqrt{3} & -2\sqrt{3} \end{pmatrix}.$$

(b) For $(x, y) = (2p\pi - \frac{\pi}{3}, 2q\pi - \frac{\pi}{3})$, the Hessian is

$$\frac{1}{2} \begin{pmatrix} 2\sqrt{3} & \sqrt{3} \\ \sqrt{3} & 2\sqrt{3} \end{pmatrix},$$

which is positive definite, so these points are local minima.

(c) For $(x, y) = (\pi + 2m\pi, \pi + 2n\pi)$, we have $f(x, y) = 0$. Moreover, consider $f(x + t, y + t)$:

$$f(x + t, y + t) = 2 \sin t (\cos t - 1).$$

Observe that the sign of the function value changes at $t = 0$. Therefore these points are saddle points.

13. $\text{grad } f = (6x^2 + 6y^2 - 12, 12xy - 12)$, so the critical points are $\pm(1, 1)$. Computing the Hessian shows that the second-derivative test is inconclusive in both cases. However,

$$f(1 + t, 1 - t) = 8t^3 - 16, \quad f(-1 + t, -1 - t) = 8t^3 + 16,$$

and at $t = 0$ neither is a maximum nor a minimum. Hence both critical points are saddle points of the original function.

□

Problem 11.6.3. Compute the maximal, minimal value of f under the given constraint if they exist.

1. $f(x, y) = xy$, $x + y = 1$
2. $f(x, y) = \cos^2 x + \cos^2 y$, $x + y = \pi/4$
3. $f(x, y) = x - 3y$, $x^2 + y^2 = 1$
4. $f(x, y) = xy$, $2x + y \leq 10, x \geq 0, y \geq 0$
5. $f(x, y) = x + y$, $x^2 + y^2 = 1$
6. $f(x, y) = x - y$, $x^2 + y^2 = 2$
7. $f(x, y) = 3x + 2y$, $2x^2 + 3y^2 \leq 3$
8. $f(x, y, z) = x^2 + 3xy + \frac{1}{2}y^2 - y - 10$, $x + y = 10$
9. $f(x, y, z) = 3x^2 + 2y^2 + z^2 + 4x + y + 3z$, $x + y + z = 4$

Proof.

Write the given function as f and constraint as g .

1. It is clear that f does not have minimal point. Therefore we may reformulate it by:

$$\begin{array}{ll} \text{maximize} & xy \\ \text{such that} & x + y = 1 \end{array}$$

Put $y = 1 - x$, then clearly the maximal value is $\frac{1}{4}$.

2. Put $y = \frac{\pi}{4} - x$, use the formula $\cos^2 t = \frac{\cos 2t + 1}{2}$, and use trigonometric identity. Then we get $f(x, \frac{\pi}{4} - x) = 1 + \frac{1}{\sqrt{2}} \sin(2x + \frac{\pi}{4})$. Therefore we get maximal value $1 + \frac{1}{\sqrt{2}}$ and minimal value $1 - \frac{1}{\sqrt{2}}$.
3. f is continuous and the domain is closed and bounded, hence both of max and min exist. Apply the Lagrangian, then we get $\text{grad } f = (1, -3)$ and $\text{grad } g = (2x, 2y)$. So we get $(x, y) = \pm \frac{1}{\sqrt{10}}(-1, 3)$, getting $\pm\sqrt{10}$ as the maximal and minimal values each.

4. Continuous, bounded, hence both maximal and minimal exist. Clearly the given function is non-negative, hence 0 would be the minimal value if we can attain such value, and clearly we can by putting 0.

Now we need to get the maximal value. Suppose that the maximal point is an interior element of the given constraint, i.e. $\text{grad } f$ should be zero. But we cannot attain that value in the interior.

Now we may apply the Lagrangian on the boundary. We may ignore boundaries $x = 0$ and $y = 0$ which give us the minimal value. Consider the boundary $2x + y = 10$. $\text{grad } f = (y, x)$ and $\text{grad } g = (2, 1)$ implies that we have $(x, y) = (\frac{5}{2}, 5)$ as a candidate. Indeed, this candidate is the maximal point, hence we have a maximal value $\frac{25}{2}$.

5. Do same thing with 3. $\pm\sqrt{2}$ are the values we want.
6. Do same thing with 3. ± 2 are the values we want.
7. $\text{grad } f$ cannot be zero, hence we must attain the maximal and minimal point on the boundary. Clearly we can attain both of them since the function is continuous and the constraint is bounded. We have $\text{grad } f = (3, 2)$ and $\text{grad } g = (4x, 6y)$, and we need to calculate $(3, 2) = \lambda(4x, 6y)$. Putting $\lambda 4x = 3$ and $\lambda 6y = 2$ into $2x^2 + 3y^2 = 3$, we get $\lambda^2 = \frac{72}{35}$. Hence we get maximal and minimal value $\pm \frac{\sqrt{70}}{2}$ each.
8. Put $y = 10 - x$, then the problem becomes the min/max problem of an one variable function. One can observe that the minimal value doesn't exist and the maximal value is $\frac{207}{2}$.
9. Put $z = 4 - x - y$, then the problem becomes the min/max problem of a two variable function. Note that the domain is unbounded, and its hessian is positive definite. Therefore we can only attain minimal value.

Finally $\text{grad } f = (6x + 4, 4y + 1, 2z + 3)$ and $\text{grad } g = (1, 1, 1)$. Therefore we get the minimal value $\frac{75}{4}$.

□

Problem 11.6.9. Compute the distance between $(1, 2, 0)$ and $z^2 = x^2 + y^2$.

Proof. We may write the problem as

$$\begin{aligned} &\text{minimize} \quad (x-1)^2 + (y-2)^2 + z^2 \\ &\text{such that} \quad x^2 + y^2 - z^2 = 0 \end{aligned}$$

One may clearly observe that this problem is equivalent to the problem minimizing $f(x, y) = (x-1)^2 + (y-2)^2 + (x^2 + y^2)$ without any constraints. Then the minimal value is $\frac{5}{2}$, hence the distance we want is $\sqrt{\frac{5}{2}}$. $\square$

Problem 11.6.15. Let $x, y, z \geq 0$ satisfy $x + y + z = 1$. Compute the maximal and minimal value of $xy + yz + zx - 2xyz$.

Proof. Put $z = 1 - (x + y)$, then we may compute the maximal/minimal value of $f(x, y) = xy + y(1 - x - y) + (1 - x - y)x - 2xy(1 - x - y)$ with constraints $x, y \geq 0$, and $x + y \leq 1$. Observe that $f(0, 0) = 0$, $f(0, y) = y(1 - y)$, $f(x, 0) = x(1 - x)$, and $\text{grad } f = ((1 - 2y)(1 - 2x - y), (1 - 2x)(1 - x - 2y))$. The first three conditions implies that we can attain the minimal value 0 and the maximal value $\frac{1}{4}$ on the boundary. Now consider the interior. Then we need $\text{grad } f = 0$, hence we have four cases:

1. $x = y = \frac{1}{2}$,
2. $y = \frac{1}{2}$, and $x = 0$,
3. $x = \frac{1}{2}$, and $y = 0$,
4. $x = y = \frac{1}{3}$.

We may only consider the first and final cases. At the first point, we have $f(\frac{1}{2}, \frac{1}{2}) = \frac{1}{4}$, and for the last point, $f(\frac{1}{3}, \frac{1}{3}) = \frac{7}{27}$. Therefore we got the maximal value $\frac{7}{27}$.² $\square$

² Indeed, the hessian matrix is negative definite at $(1/3, 1/3)$.

Problem 12.1.2. *Let*

$$F(x, y) = (x^2 - y^2, 2xy).$$

Find the Jacobian matrix and the Jacobian determinant. Also, let

$$G(x, y) = (x^3 - xy^2, x^2y - y^3).$$

Find the Jacobian matrix and the Jacobian determinant.

Proof.

1. For $F(x, y) = (x^2 - y^2, 2xy)$,

$$F'(x, y) = \begin{pmatrix} 2x & -2y \\ 2y & 2x \end{pmatrix}, \quad \det F'(x, y) = 4(x^2 + y^2).$$

2. For $G(x, y) = (x^3 - xy^2, x^2y - y^3)$,

$$F'(x, y) = \begin{pmatrix} 3x^2 - y^2 & -2xy \\ 2xy & x^2 - 3y^2 \end{pmatrix}, \quad \det F'(x, y) = 3x^4 - 6x^2y^2 + 3y^4.$$

□

Problem 12.1.3. At the given point P , find the Jacobian matrix of each function.

1. $F(x, y) = (x^2 + y^2, x^2 - y^2), \quad P = (1, 2).$

2. $F(x, y) = (x + y, x - y, xy), \quad P = (1, 2).$

3. $F(x, y) = (\sin x, \cos(xy)), \quad P = (\pi, 1).$

4. $F(x, y, t) = (x, y), \quad P = (1, 1, 1).$

Proof. 1.

$$F'(x, y) = \begin{pmatrix} 2x & 2y \\ 2x & -2y \end{pmatrix}, \quad F'(1, 2) = \begin{pmatrix} 2 & 4 \\ 2 & -4 \end{pmatrix}.$$

2.

$$F'(x, y) = \begin{pmatrix} 1 & 1 \\ 1 & -1 \\ y & x \end{pmatrix}, \quad F'(1, 2) = \begin{pmatrix} 1 & 1 \\ 1 & -1 \\ 2 & 1 \end{pmatrix}.$$

3.

$$F'(x, y) = \begin{pmatrix} \cos x & 0 \\ -y \sin(xy) & -x \sin(xy) \end{pmatrix}, \quad F'(\pi, 1) = \begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix}.$$

4.

$$F'(x, y, t) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad F'(1, 1, 1) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

□

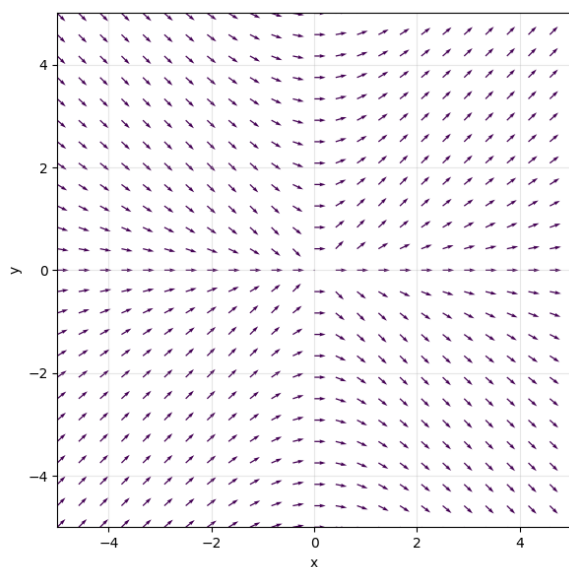
Problem 13.1.1. Draw the following vector fields.

1. $\mathbf{F}(x, y) = (x^2 - y^2) \mathbf{i} + 2xy \mathbf{j}$.

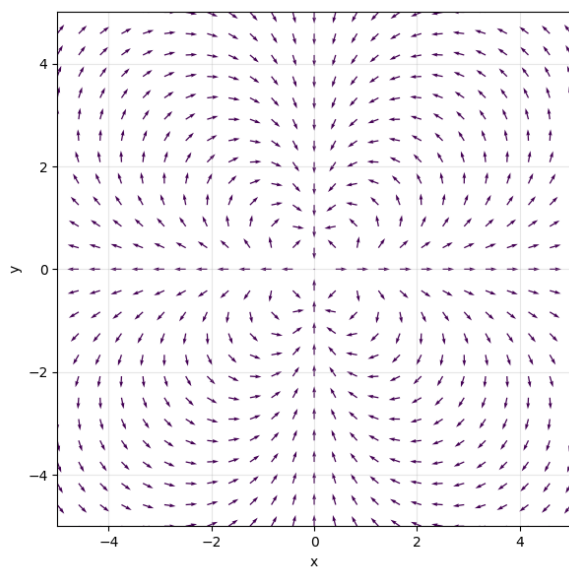
2. $\mathbf{F}(x, y) = (x^3 - 3xy^2) \mathbf{i} + (3x^2y - y^3) \mathbf{j}$.

3. $\mathbf{F}(x, y, z) = -y \mathbf{i} + x \mathbf{j}$.

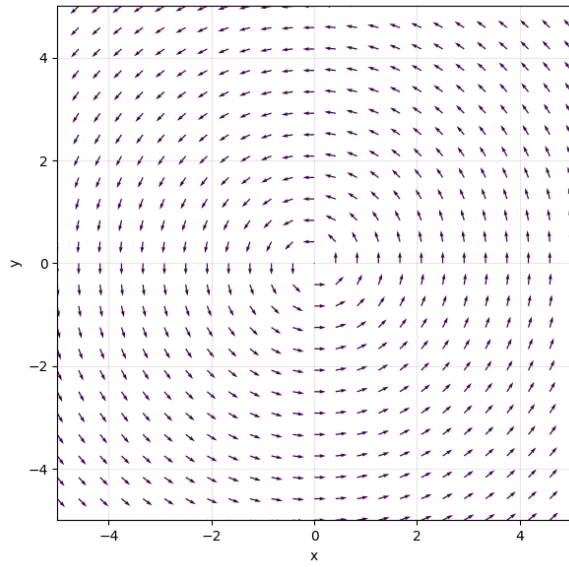
Proof.



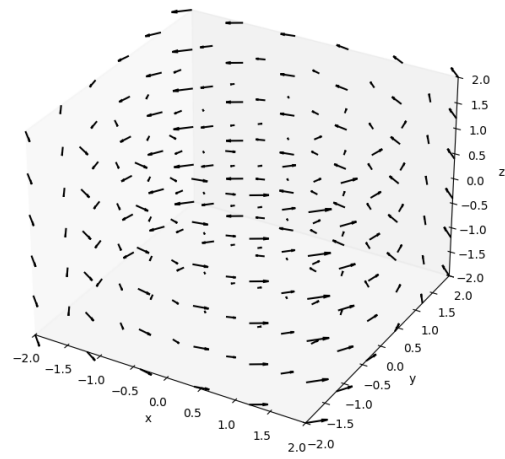
1.



2.



3.



□

Problem 13.1.6. Let the *angular momentum* of a moving point $X(t)$ of mass m about the origin be

$$\mathbf{L}(t) := X(t) \times (mX'(t))$$

1. Show that any curve whose angular momentum is a nonzero constant vector is a plane curve.
2. Suppose a curve $X(t)$ satisfies equation $\mathbf{F}(X(t)) = mX''(t)$, and that the vector field $\mathbf{F}$ at each point is proportional to the position vector field. Prove that the angular momentum is a constant vector.

Proof.

1.

$$\begin{aligned} X(t) \cdot \mathbf{L}(t) &= \det(X(t), X(t), mX'(t)) = 0 \\ \Rightarrow X(t) \cdot \mathbf{L}(t) &= X(t) \cdot \mathbf{a} = 0 \quad \text{for some constant } \mathbf{a}. \end{aligned}$$

2. We may write $mX''(t) = kX(t)$. Then

$$\frac{d}{dt}\mathbf{L}(t) = X'(t) \times (mX'(t)) + X(t) \times (mX''(t)) = 0$$

□

Problem 13.2.1. Integrate the vector field $\mathbf{r}^*(x, y) = (-y, x)$ along the following curves:

1. $X(t) = (\cos t, \sin t), \quad 0 \leq t \leq 4\pi$
2. $X(t) = (\cos t, -\sin t), \quad 0 \leq t \leq 2\pi$
3. $X : (0, 0) \rightarrow (3, 0) \rightarrow (0, 3) \rightarrow (0, 0)$

Proof.

1. $\int_X \mathbf{r} \cdot d\mathbf{s} = \int_0^{4\pi} (-\sin t, \cos t) \cdot (-\sin t, \cos t) dt = 4\pi.$
2. $\int_X \mathbf{r} \cdot d\mathbf{s} = \int_0^{2\pi} (\sin t, \cos t) \cdot (-\sin t, -\cos t) dt = -2\pi.$
3. Parametrize the curve as follows:

$$X(t) = \begin{cases} (3t, 0), & 0 \leq t \leq 1, \\ (6 - 3t, 3t - 3), & 1 \leq t \leq 2, \\ (0, 9 - 3t), & 2 \leq t \leq 3. \end{cases}$$

$$\int_X \mathbf{r} \cdot d\mathbf{s} = \int_0^1 (0, 3t) \cdot (3, 0) dt + \int_1^2 (3 - 3t, 6 - 3t) \cdot (-3, 3) dt + \int_2^3 (3t - 9, 0) \cdot (0, -3) dt = 9.$$

□

Problem 13.2.3. Let $X : [t_0, t_1] \rightarrow \mathbb{R}^2 - \{O\}$ be a curve that does not pass through the origin. There exists a continuous function $\theta(t)$ such that

$$X(t) = |X(t)|(\cos \theta(t), \sin \theta(t))$$

We then define

$$\text{wind}(X) := \frac{1}{2\pi}(\theta(t_1) - \theta(t_0))$$

to be the **winding number** of the curve X about the origin.

1. Compute the winding number of the following curves about the origin.

(a) $X(t) := (\cos 3t, \sin 3t), \quad 0 \leq t \leq 2\pi.$

(b) $Y(t) := (\cos t, \sin 3t), \quad 0 \leq t \leq 2\pi.$

2. Explain why the winding number is an integer when X is a closed curve.

3. If X is a differentiable curve, show that

$$\text{wind}(X) = \frac{1}{2\pi} \int_X \mathbf{a} \cdot d\mathbf{s},$$

where $\mathbf{a}$ denotes the angular vector field.

Proof.

1. (a) 3

(b) Even though we may just consider the given graph in the book, I don't like that solution, because it is not rigorous. We solve this by using the subproblem 3 below. Since $Y(t)$ is differentiable, we may apply it. Then, we get:

$$\begin{aligned} \text{wind}(Y) &= \frac{1}{2\pi} \int_Y \mathbf{a} \cdot d\mathbf{s} = \frac{1}{2\pi} \int_0^{2\pi} \frac{(-\sin 3t, \cos t)}{\sin^2 3t + \cos^2 t} \cdot (-\sin t, 3 \cos 3t) dt \\ &= \frac{1}{2\pi} \int_0^{2\pi} \frac{\sin t \sin 3t + 3 \cos t \cos 3t}{\sin^2 3t + \cos^2 t} dt \end{aligned}$$

Observe that $-\frac{d}{dt} \arctan\left(\frac{\cos t}{\sin 3t}\right) = \frac{\sin t \sin 3t + 3 \cos t \cos 3t}{\sin^2 3t + \cos^2 t}$. But since this function is discontinuous at several points, we need to "regularize" it by defining:³

³ or we may just integrate by dividing intervals.

$$f(t) = \begin{cases} \frac{\pi}{2} & t = 0 \\ \arctan\left(\frac{\cos t}{\sin 3t}\right) & 0 < t < \frac{\pi}{3} \\ \frac{\pi}{2} & t = \frac{\pi}{3} \\ \pi + \arctan\left(\frac{\cos t}{\sin 3t}\right) + \pi & \frac{\pi}{3} < t < \frac{2\pi}{3} \\ \frac{3\pi}{2} & t = \frac{2\pi}{3} \\ \arctan\left(\frac{\cos t}{\sin 3t}\right) + 2\pi & \frac{2\pi}{3} < t < \pi \\ \frac{3\pi}{2} & t = \pi \\ \arctan\left(\frac{\cos t}{\sin 3t}\right) + \pi & \pi < t < \frac{4\pi}{3} \\ \frac{3\pi}{2} & t = \frac{4\pi}{3} \\ \arctan\left(\frac{\cos t}{\sin 3t}\right) + 2\pi & \frac{4\pi}{3} < t < \frac{5\pi}{3} \\ \frac{5\pi}{2} & t = \frac{5\pi}{3} \\ \arctan\left(\frac{\cos t}{\sin 3t}\right) + 3\pi & \frac{5\pi}{3} < t < 2\pi \\ \frac{5\pi}{2} & t = 2\pi \end{cases}$$

Then finally we get:

$$\begin{aligned} \text{wind}(Y) &= \frac{1}{2\pi} \int_0^{2\pi} \frac{\sin t \sin 3t + 3 \cos t \cos 3t}{\sin^2 3t + \cos^2 t} dt \\ &= -\frac{1}{2\pi} \int_0^{2\pi} f'(t) dt \\ &= -2\pi \end{aligned}$$

Therefore the winding number is -1 .

2. If X is closed, then $\theta(t_1) - \theta(t_0) = 2n\pi$ where $n \in \mathbb{Z}$.
3. Since X does not meet the origin, we may write $X(t) = r(t)(\cos \theta(t), \sin \theta(t))$ where $r(t) = |X(t)|$ where r and θ are C^1 . Then:

$$\begin{aligned} & \int_X \mathbf{a} \cdot ds \\ &= \int_{t_0}^{t_1} \frac{r(-\sin \theta, \cos \theta)}{r^2} \cdot \left(\frac{dr}{dt}(\cos \theta, \sin \theta) + r(-\sin \theta, \cos \theta) \frac{d\theta}{dt} \right) dt \\ &= \int_{t_0}^{t_1} \frac{d\theta}{dt} dt \\ &= \theta(t_1) - \theta(t_0) \end{aligned}$$

□

Problem 13.2.4. Find the winding number about the origin of each planar curve below.

1. $X(t) = (2 + \cos 3t)(\cos 2t, \sin 2t), \quad 0 \leq t \leq 2\pi.$
2. $X(t) = (2 + \cos 4t)(\cos 2t, \sin 2t), \quad 0 \leq t \leq 2\pi.$
3. $X(t) = (2 + \cos 5t)(\cos 2t, \sin 2t), \quad 0 \leq t \leq 2\pi.$
4. $X(t) = (2 + \cos 5t)(\cos 3t, \sin 3t), \quad 0 \leq t \leq 2\pi.$

Also determine, for a ray starting at the origin, the number of intersection points of each curve with that ray.

Proof. Finding number of self-intersection of curve $X(t) = (2 + \cos kt)(\cos mt, \sin mt)$ is equivalent to the given problem. You may solve this rigorously by finding solution of $X(t_1) = X(t_0)$, i.e. $kt_1 = kt_0 + \frac{2n\pi}{k}$ and $mt_1 = mt_0 + \frac{2n\pi}{m}$.

1. 2, number of intersections are 2 except for 3 rays which intersects only once.
2. 2, number of intersections are all 1.
3. 2, number of intersections are 2 except for 5 rays which intersects only once.
4. 3, number of intersections are 3 except for 10 rays which intersects only twice.

□

Problem 13.3.3. Consider the vector field $\mathbf{F}(x, y) = (-y, x)$ and the curve

$$X(t) := (\cos t, \sin t), \quad 0 \leq t \leq \pi.$$

Compute the line integral $\int_X \mathbf{F} \cdot d\mathbf{s}$. Also define

$$\tilde{X}(t) := (\cos 3t, \sin 3t), \quad 0 \leq t \leq \pi,$$

and show that although the two curves $X, \tilde{X} : [0, \pi] \rightarrow \mathbb{R}^2$ have the same boundary,

$$\int_{\tilde{X}} \mathbf{F} \cdot d\mathbf{s} \neq \int_X \mathbf{F} \cdot d\mathbf{s}.$$

Proof.

$$\int_X \mathbf{F} \cdot d\mathbf{s} = \int_0^\pi (-\sin t, \cos t) \cdot (-\sin t, \cos t) dt = \pi,$$

$$\int_{\tilde{X}} \mathbf{F} \cdot d\mathbf{s} = \int_0^\pi (-\sin 3t, \cos 3t) \cdot (-3\sin 3t, 3\cos 3t) dt = 3\pi.$$

Since $X(0) = (1, 0) = \tilde{X}(0)$ and $X(\pi) = (-1, 0) = \tilde{X}(\pi)$, the two curves have the same boundary, but $\int_{\tilde{X}} \mathbf{F} \cdot d\mathbf{s} \neq \int_X \mathbf{F} \cdot d\mathbf{s}$. □

Problem 13.3.7. Find a potential function of the following vector fields.

1. $\mathbf{F}(x, y, z) = (yz, zx, xy)$
2. $\mathbf{F}(x, y, z) = (2x - y - z, 2y - x, 2z - x)$
3. $\mathbf{F}(x, y, z) = (y \cos z - yze^x, x \cos z - ze^x, -xy \sin z - ye^x)$
4. $\mathbf{F}(x, y, z) = (yz, xz + y, xy + 1)$

Proof. By the uniqueness of the potential function, we suffice to find only one such function.

1. $xyz + C$
2. $x^2 + y^2 + z^2 - xy - xz + C$
3. $xy \cos z - yze^x + C$
4. $xyz + \frac{1}{2}y^2 + z + C$

□

Problem 13.3.10. In the xyz -space, consider the vector field defined on the region excluding the z -axis

$$\mathbf{F} = \left(\frac{x}{x^2 + y^2} + y \cos(xy + z) \right) \mathbf{i} + \left(\frac{y}{x^2 + y^2} + x \cos(xy + z) \right) \mathbf{j} + \cos(xy + z) \mathbf{k}.$$

1. If $\mathbf{F}$ has potential functions, find all of them.
2. For the curve

$$X(t) = (\cos t, \sin t, t), \quad 0 \leq t \leq \frac{\pi}{2},$$

compute the line integral $\int_X \mathbf{F} \cdot d\mathbf{s}$.

Proof.

1. $\varphi(x, y, z) = \frac{1}{2} \log(x^2 + y^2) + \sin(xy + z) + C$

- 2.

$$\int_X \mathbf{F} \cdot d\mathbf{s} = \varphi\left(X\left(\frac{\pi}{2}\right)\right) - \varphi(X(0)) = 1.$$

□

Problem 13.3.12. In the space with the z -axis removed, consider the vector field

$$\mathbf{a}(x, y, z) = -\frac{y}{x^2 + y^2} \mathbf{i} + \frac{x}{x^2 + y^2} \mathbf{j}.$$

Is this vector field closed? Does this vector field admit a potential function on its entire domain? Does it admit a potential function locally?

Proof. Write as:

$$\mathbf{a}(x, y, z) = \left(-\frac{y}{x^2 + y^2}, \frac{x}{x^2 + y^2}, 0 \right) = (f_1(x, y, z), f_2(x, y, z), f_3(x, y, z)).$$

Then

$$\frac{\partial f_1}{\partial y} = \frac{y^2 - x^2}{(x^2 + y^2)^2} = \frac{\partial f_2}{\partial x}, \quad \frac{\partial f_2}{\partial z} = 0 = \frac{\partial f_3}{\partial y}, \quad \frac{\partial f_1}{\partial z} = 0 = \frac{\partial f_3}{\partial x}.$$

Hence the given vector field is closed.

But consider the curve $X(t) = (\cos t, \sin t, 0)$, $0 \leq t \leq 2\pi$. Then

$$\int_X \mathbf{a} \cdot d\mathbf{s} \neq 0,$$

so the given vector field does not have a potential function. Nevertheless, by the Poincaré lemma, it has a potential function locally. $\square$

Problem 13.4.1. Compute the following integrals.

1. $\int_X xy \, dx + (x + y) \, dy$, X is the portion of the curve $y = x^2$ from $(-1, 1)$ to $(2, 4)$.
2. $\int_X y^2 \, dx + x \, dy$, X is the portion of the curve $x = y^3$ from $(-1, -1)$ to $(1, 1)$.
3. $\int_X y\sqrt{x} \, dx + x\sqrt[3]{y} \, dy$, X is the portion of the curve $y^2 = x^3$ from $(1, 1)$ to $(4, 8)$.
4. $\int_X x^2y \, dx + xy^3 \, dy$, X is the broken line from $(-1, 1)$ to $(2, 1)$, and from $(2, 1)$ to $(2, 5)$.
5. $\int_X (x + 2y) \, dx + (2x - y) \, dy$, X is the broken line from $(3, 2)$ to $(3, -1)$, and from $(3, -1)$ to $(-2, -1)$.

Proof. 1. The given curve is parameterized by $X(t) = (t, t^2)$, $-1 \leq t \leq 2$.

$$\int_X xy \, dx + (x + y) \, dy = \int_{-1}^2 (t^3, t^2 + t) \cdot (1, 2t) \, dt = \frac{69}{4}.$$

2. The given curve is parameterized by $X(t) = (t^3, t)$, $-1 \leq t \leq 1$.

$$\int_X y^2 \, dx + x \, dy = \int_{-1}^1 (t^2, t^3) \cdot (3t^2, 1) \, dt = \frac{6}{5}.$$

3. The given curve is parameterized by $X(t) = (t^2, t^3)$, $1 \leq t \leq 2$.

$$\int_X y\sqrt{x} \, dx + x\sqrt[3]{y} \, dy = \int_1^2 (t^4, t^3) \cdot (2t, 3t^2) \, dt = \frac{105}{2}.$$

4. Let $X_1(t) = (t, 1)$, $-1 \leq t \leq 2$, and $X_2(t) = (2, t)$, $1 \leq t \leq 5$. Then $X = X_1 + X_2$.

$$\int_X x^2y \, dx + xy^3 \, dy = \int_{X_1} x^2y \, dx + xy^3 \, dy + \int_{X_2} x^2y \, dx + xy^3 \, dy = \int_{-1}^2 (t^2, t) \cdot (1, 0) \, dt + \int_1^5 (4t, 2t^3) \cdot (0, 1) \, dt = 315.$$

5. Let $X_1(t) = (3, -t)$, $-2 \leq t \leq 1$, and $X_2(t) = (-t, -1)$, $-3 \leq t \leq 2$. Then

$X = X_1 * X_2$, hence we get

$$\begin{aligned} \int_X (x + 2y) \, dx + (2x - y) \, dy &= \int_{X_1} (x + 2y) \, dx + (2x - y) \, dy + \int_{X_2} (x + 2y) \, dx + (2x - y) \, dy \\ &= \int_{-2}^1 (3 - 2t, 6 + t) \cdot (0, -1) \, dt + \int_{-3}^2 (-t - 2, -2t + 1) \cdot (-1, 0) \, dt \\ &= -9. \end{aligned}$$

□

3

Multi-Integrals and Green's Theorem

3.1 problem

Problem 14.2.1. Show that triangles have area. Also show that any polyhedrons and disks have area.

Proof. Fixing one edge of triangle on the x -axis, we may write the other edges as lines on xy -plane. Since any lines are integrable, by the theorem 2.0.1, we may say that triangles are integrable. By dividing polyhedrons into several triangles, we may say that polyhedrons have area. Finally, we may assume that the center of disk is origin. Then we may write the upper semi-circle as $y = r\sqrt{1 - x^2}$, which is integrable. $\square$

Problem 14.2.2 (Mean Value Theorem). Let $A \subseteq \mathbb{R}^n$ be a path connected set. Also, let the function $f : A \rightarrow \mathbb{R}$ be continuous and bounded. Put $\bar{f}$ be the mean value of f . Then there exists $p \in A$ such that $f(p) = \bar{f}$.

Proof. Let $m = \inf f$ and $M = \sup f$.¹ If $\bar{f} = m$ or M , then $f \equiv m$ or M , hence we are done. Now suppose not, i.e. $m < \bar{f} < M$. Then there exist $p_0, p_1 \in A$ such that $m < f(p_0) < \bar{f} < f(p_1) < M$. Since A is path connected, we may take a continuous function $\gamma : I \rightarrow A$ such that $\gamma(0) = p_0$ and $\gamma(1) = p_1$. Then by the IVT, there exists $t_0 \in I$ such that $f(\gamma(t_0)) = \bar{f}$, where $\gamma(t_0) \in A$. $\square$

¹ You may not use min or max, because they may not exist.

Problem 14.2.6. Let $A \subseteq \mathbb{R}^n$ has $h(x, y, z)$ as its density function. Compute its mean density.

Proof. By the definition,

$$\frac{1}{\text{vol}(A)} \int_A h(x, y, z) \, dx dy dz$$

□

4

Green's Theorem

15.1 problem

Problem 15.5.2. Calculate the following integrals by using Green's Theorem.

1. $\int_C (x + y^2) dx + (y + x^2) dy,$

C is the boundary of the square with vertices $(\pm 1, \pm 1)$

2. $\int_C (x^2 + y^2) dx - 2xy dy,$

C is the boundary of the triangle enclosed by $x = 0$, $y = 0$, and $x + y = 1$

3. $\int_C (y + e^x) dx + (2x^2 + \cos y) dy,$

C is the boundary of the triangle with vertices $(0, 0)$, $(1, 1)$, and $(2, 0)$

4. $\int_C (x^2 - y^2) dx + xy dy,$

C is the boundary of the region enclosed by $y = x$ and $y = x^2$

5. $\int_C (-y^2 + e^x) dx + (\arctan y) dy,$

C is the boundary of the region enclosed by $y = x^2$ and $x = y^2$

6. $\int_C x^2 dy - y^2 dx,$

C is the curve given in polar coordinates by $r = 1 + \cos \theta$

Proof. Apply Green's theorem (Stokes theorem in the special case):

$$\int_{\partial\Omega} \omega = \int_{\Omega} d\omega$$

1. $\int_C (x + y^2) dx + (y + x^2) dy = \iint_D 2x - 2y dx \wedge dy = 0$

2. $\int_C (x^2 + y^2) dx - 2xy dy = \iint_D -4y dx \wedge dy = \int_0^1 \int_0^{1-x} -4y dy dx = -\frac{2}{3}$
3. $\int_C (y + e^x) dx + (2x^2 + \cos y) dy = \iint_D 4x - 1 dx \wedge dy = \int_0^1 \int_0^x 4x - 1 dy dx + \int_1^2 \int_0^{2-x} 4x - 1 dy dx = 3$
4. $\int_C (x^2 - y^2) dx + xy dy = \iint_D 3y dx \wedge dy = \int_0^1 \int_{x^2}^x 3y dy dx = \frac{1}{5}$
5. $\int_C (-y^2 + e^x) dx + (\arctan y) dy = \iint_D 2y dx \wedge dy = \int_0^1 \int_{x^2}^{\sqrt{x}} 2y dy dx = \frac{3}{10}$
6. $\int_C x^2 dy - y^2 dx = \iint_D 2x + 2y dx \wedge dy = \int_0^{2\pi} \int_0^{1+\cos\theta} 2r^2 (\cos\theta + \sin\theta) dr d\theta = \int_0^{2\pi} \frac{2}{3} (1 + \cos\theta)^3 (\cos\theta + \sin\theta) d\theta = \frac{5\pi}{2}$

□

Problem 15.5.5. Let the center of D be $(\bar{x}, \bar{y})$. Put $A = \text{Area}(D)$, and assume that ∂D is piecewise C^1 . Show that

$$\bar{x} = \frac{1}{2A} \int_{\partial D} x^2 dy, \quad \bar{y} = -\frac{1}{2A} \int_{\partial D} y^2 dx.$$

Proof.

$$\frac{1}{2A} \int_{\partial D} x^2 dy = \frac{1}{2A} \int_D 2x dx \wedge dy = \bar{x}$$

Same for y .

□

Problem 15.5.7. Let C be a simple closed curve, and $D = \text{int}(C)$ contain the origin.

Then show that:

$$\int_C \mathbf{a} \cdot d\mathbf{s} = 2\pi, \quad \iint_D \text{rot } \mathbf{a} dx \wedge dy = 0$$

where $\mathbf{a} = \left(-\frac{y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right)$. Conclude by saying why this is compatible with the Green's theorem.

Proof. Integrations can be easily shown by simple calculation. This is compatible with the Green's theorem since $\mathbf{a}$ is not defined globally in D .

□

5

Divergence Theorem and Stokes' Theorem

16.1 problem

Problem 16.1.1. Parametrize $z = x^2 + y^2$.

Proof. $(x, y, x^2 + y^2), (r \cos \theta, r \sin \theta, r^2) \dots$

□

Problem 16.1.2. Parametrize the following surfaces.

1. $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

2. $z = x^2 + 2y^2$

3. $x^2 + 2y^2 - 3z^2 = 0$

4. $x^2 + 2y^2 - 3z^2 = 1$

5. $x^2 - 2y^2 - 3z^2 = 1$

Proof. Nothing special for (1)-(3). I will only state the answers.

1. $(a \sin u \cos t, b \sin u \sin t, c \cos u), \quad 0 \leq u \leq \pi, 0 \leq t \leq 2\pi$

2. $(\sqrt{2}u \cos v, u \sin v, 2u^2), \quad u \geq 0, 0 \leq v \leq 2\pi$

3. $(\sqrt{6}u \cos t, \sqrt{3}u \sin t, \sqrt{2}u), \quad u \in \mathbb{R}, 0 \leq t \leq 2\pi$

For (4) and (5), one may just 'memorize' how to parametrize it... by using hyperbolic functions, for using the equation $\sec^2 t - \tan^2 t = 1$. But here, let me introduce (maybe) a better way. Our main problem is that there is a minus sign in front of squared variable(s). But well, you may know that $(i)^2 = -1$. So we will use that. Note that this is based on the analytic continuity of functions.

4. Since we may just put constant terms later, we only suffice to parametrize $x^2 + y^2 - z^2 = 1$. Put $X = x$, $Y = y$, and $Z = iz$. Then one gets:

$$X^2 + Y^2 + Z^2 = 1 \Rightarrow (X, Y, Z) = (\cos u \cos t, \cos u \sin t, \sin u)$$

Note that we should put $\sin u$ at the position where i will work on. This is because we will use the following equations:

$$\cosh(v) = \cos(iv)$$

$$i \sinh(v) = \sin(iv)$$

This equations are based on the **Euler equation**. Now put $X = x$, $Y = y$, and $Z = iz$. Note that as soon as we put as such, u and t don't need to be real numbers, as we extended the domain to complex numbers, $\mathbb{C}$. So we get:

$$(x, y, z) = (\cos u \cos t, \cos u \sin t, -i \sin u)$$

But we should make $(x, y, z) \in \mathbb{R}$, i.e. we need to make z real. To do so, we need to put appropriate complex numbers. And here is where our equations should work on! Put $u = iv$. Then one may observe that:

$$\begin{aligned} (x, y, z) &= (\cos(iv) \cos t, \cos(iv) \sin t, -i \sin(iv)) \\ &= (\cosh v \cos t, \cosh v \sin t, \sinh v) \end{aligned}$$

This is the parametrization what we were finding for! Indeed, you may check this parametrization works. Here we may note that we considered sign of parametrization. In this problem we have no issues, but in the next problem, you will understand what I am talking about. Now to conclude, we should attach constant terms. The final answer is:

$$(x, y, z) = (\cosh v \cos t, \frac{1}{\sqrt{2}} \cosh v \sin t, \frac{1}{\sqrt{3}} \sinh v)$$

5. We use the exactly same method. We suffice to parametrize $x^2 - y^2 - z^2 = 1$. Put $X = x$, $Y = iy$, and $Z = iz$. Then we get:

$$(X, Y, Z) = (\cos u, \sin u \cos t, \sin u \sin t)$$

Again, note that we should put $\sin$ at the place where i will work on. Therefore we get:

$$\begin{aligned}(x, y, z) &= (\cos u, -i \sin u \cos t, -i \sin u \sin t) \quad \dots \text{Put } u = iv \\ &= (\cos(iv), -i \sin(iv) \cos t, -i \sin(iv) \sin t) \\ &= (\cosh v, \sinh v \cos t, \sinh v \sin t)\end{aligned}$$

But you should note that we **didn't** consider the case where $x < 0$. As you can see, in this parametrization, x is always positive. But here is a simple solution: just take another parametrization. Therefore we get two-chart parametrization, attaching constant terms:

$$(x, y, z) = \begin{cases} (\cosh v, \frac{1}{\sqrt{2}} \sinh v \cos t, \frac{1}{\sqrt{3}} \sinh v \sin t) & x > 0 \\ (-\cosh v, \frac{1}{\sqrt{2}} \sinh v \cos t, \frac{1}{\sqrt{3}} \sinh v \sin t) & x < 0 \end{cases}$$

Splitting the parametrization is unavoidable since the surface has two connected complements.

Finally, when you write the answer of your exam, then do not say anything about our friend i , but rather, just say "I just observed this parametrization works! As we can see, ... (just introduce your parametrization)." $\square$

Problem 16.1.3. Parametrize the torus obtained by rotating a circle in the yz -plane centered at $(0, R, 0)$ with radius r about the z -axis. (Assume $0 < r < R$.)

Proof. Easy to do:

$$\left((R + r \cos t) \cos \theta, (R + r \cos t) \sin \theta, r \sin t \right).$$

Moreover, we can also represent the torus as follows:

$$(\sqrt{x^2 + y^2} - R)^2 + z^2 = r^2.$$

$\square$

Problem 16.2.7. Among the parts of the sphere $x^2 + y^2 + z^2 = 1$, find the area of the portion lying inside the elliptic cylinder $x^2 + \frac{y^2}{a^2} \leq 1$. (Here $0 < a \leq 1$.)

Proof. Parametrize the surface by

$$X(x, y) = (x, y, \sqrt{1 - x^2 - y^2}), \quad x^2 + \frac{y^2}{a^2} \leq 1.$$

Now

$$X_x = \left(1, 0, \frac{-x}{\sqrt{1 - x^2 - y^2}}\right), \quad X_y = \left(0, 1, \frac{-y}{\sqrt{1 - x^2 - y^2}}\right),$$

therefore

$$|X_x \times X_y| = \left| \left(\frac{x}{\sqrt{1 - x^2 - y^2}}, \frac{y}{\sqrt{1 - x^2 - y^2}}, 1 \right) \right| = \frac{1}{\sqrt{1 - x^2 - y^2}}.$$

Thus, by symmetry, the area of the desired part is

$$\begin{aligned} 2 \int_{-1}^1 \int_{-a\sqrt{1-x^2}}^{a\sqrt{1-x^2}} \frac{1}{\sqrt{1 - x^2 - y^2}} dy dx &= \int_{-1}^1 \int_{-\arcsin a}^{\arcsin a} \frac{\sqrt{1 - x^2} \cos t}{\sqrt{1 - x^2} \cos t} dt dx \\ &= 8 \sin^{-1} a, \end{aligned}$$

by the substitution $y = \sqrt{1 - x^2} \sin t$.

□

Problem 16.2.8. For $a \in \mathbb{R}$, let $z \geq a\sqrt{x^2 + y^2}$ represent a cone. Show that the solid angle at its vertex O is

$$2\pi \left(1 - \frac{a}{\sqrt{1 + a^2}} \right).$$

Proof. If $a = 0$, ok.

If $a > 0$, if we express the given region in spherical coordinates, we obtain

$$\varphi \leq \tan^{-1} \frac{1}{a}.$$

Therefore the solid angle is

$$\int_0^{2\pi} \int_0^{\tan^{-1} \frac{1}{a}} \sin \varphi d\varphi d\theta = 2\pi \left(1 - \frac{a}{\sqrt{1 + a^2}} \right).$$

Finally, if $a < 0$, just subtract the case of $a > 0$ from 4π . Therefore the result holds. □

Problem 16.3.1. On the unit sphere $S : x^2 + y^2 + z^2 = 1$, compute the integral values of

the following functions:

$$x, y, z, x^2, y^2, z^2, xy, yz, zx, x^2y, xyz$$

Proof. By the symmetry, all of integrations except for x^2 , y^2 , and z^2 are zero. But again by the symmetry, we know that $\iint_S x^2 dA = \iint_S y^2 dA = \iint_S z^2 dA$. Therefore:

$$\begin{aligned} \iint_S x^2 + y^2 + z^2 dA &= \iint_S 1 dA = 4\pi \\ \therefore \iint_S x^2 dA &= \iint_S y^2 dA = \iint_S z^2 dA = \frac{4\pi}{3}. \end{aligned}$$

□

Problem 16.3.2. Find the area and the centroid of the following surfaces. (Here $r = \sqrt{x^2 + y^2}$, $\rho = \sqrt{x^2 + y^2 + z^2}$.)

1. The paraboloid $z = r^2$, $0 \leq r \leq a$.
2. The cone $z = kr$, $0 \leq r \leq a$.
3. The part of the sphere $\rho = a$ lying inside the cone $z = r$.
4. The part of the sphere $x^2 + y^2 + z^2 = 4$ inside the cylinder $x^2 + y^2 = 2x$ with $z \geq 0$.

Proof.

1. Parametrization:

$$X(r, \theta) = (r \cos \theta, r \sin \theta, r^2), \quad 0 \leq r \leq a, \quad 0 \leq \theta \leq 2\pi.$$

Then

$$X_r = (\cos \theta, \sin \theta, 2r),$$

$$X_\theta = (-r \sin \theta, r \cos \theta, 0),$$

$$|X_r \times X_\theta| = r\sqrt{1 + 4r^2}.$$

Area:

$$\begin{aligned} \iint_S dS &= \int_0^{2\pi} \int_0^a |X_r \times X_\theta| dr d\theta = 2\pi \int_0^a r\sqrt{1 + 4r^2} dr \\ &= 2\pi \cdot \frac{1}{12} \left((1 + 4a^2)^{3/2} - 1 \right) = \frac{\pi}{6} \left((1 + 4a^2)^{3/2} - 1 \right). \end{aligned}$$

Centroid: By symmetry $\bar{x} = \bar{y} = 0$. Moreover,

$$\begin{aligned}
 \int_S z \, dS &= \int_0^{2\pi} \int_0^a r^2 |X_r \times X_\theta| \, dr \, d\theta \\
 &= \int_0^{2\pi} \int_0^a r^3 (1 + 4r^2)^{1/2} \, dr \, d\theta \\
 &= 2\pi \left(-\frac{1}{12} r^2 (1 + 4r^2)^{3/2} \Big|_0^a - \frac{1}{12} \int_0^a 2r (1 + 4r^2)^{3/2} \, dr \right) \\
 &= \frac{\pi}{6} \left(a^2 (1 + 4a^2)^{3/2} - \frac{1}{10} \left((1 + 4a^2)^{5/2} \right) \right)
 \end{aligned}$$

Therefore:

$$\bar{z} = \frac{1}{\text{area } S} \int_S z \, dS = \frac{\left(a^2 (1 + 4a^2)^{3/2} - \frac{1}{10} \left((1 + 4a^2)^{5/2} \right) \right)}{\left((1 + 4a^2)^{3/2} - 1 \right)}$$

2. Parametrization: $X(r, \theta) = (r \cos \theta, r \sin \theta, kr)$, $0 \leq r \leq a$. Therefore $X_r = (\cos \theta, \sin \theta, k)$, $X_\theta = (-r \sin \theta, r \cos \theta, 0)$, $|X_r \times X_\theta| = r\sqrt{1 + k^2}$.

Area:

$$\int_0^{2\pi} \int_0^a |X_r \times X_\theta| \, dr \, d\theta = \pi a^2 \sqrt{1 + k^2}.$$

Centroid: By symmetry $\bar{x} = \bar{y} = 0$. Moreover,

$$\int_S z \, dS = \int_0^{2\pi} \int_0^a kr |X_r \times X_\theta| \, dr \, d\theta = \int_0^{2\pi} \int_0^a kr^2 \sqrt{1 + k^2} \, dr \, d\theta = \frac{2\pi}{3} ka^3 \sqrt{1 + k^2}.$$

Therefore

$$\bar{z} = \frac{1}{\text{area } S} \int_S z \, dS = \frac{2}{3} ak.$$

3. Parametrization: $X(\theta, \varphi) = a(\sin \varphi \cos \theta, \sin \varphi \sin \theta, \cos \varphi)$, $z \geq r$ and $a \cos \varphi \geq a \sin \varphi$, hence $0 \leq \varphi \leq \frac{\pi}{4}$. Also we get: $X_\theta = a(-\sin \varphi \sin \theta, \sin \varphi \cos \theta, 0)$, $X_\varphi = a(\cos \varphi \cos \theta, \cos \varphi \sin \theta, -\sin \varphi)$, and $|X_\theta \times X_\varphi| = a^2 \sin \varphi$.

Area:

$$\int_0^{2\pi} \int_0^{\pi/4} |X_\theta \times X_\varphi| \, d\varphi \, d\theta = \int_0^{2\pi} \int_0^{\pi/4} a^2 \sin \varphi \, d\varphi \, d\theta = 2\pi a^2 \left(1 - \frac{1}{\sqrt{2}} \right).$$

Centroid: By symmetry $\bar{x} = \bar{y} = 0$. Also:

$$\begin{aligned}\int_S z \, dS &= \int_0^{2\pi} \int_0^{\pi/4} a \cos \varphi |X_\theta \times X_\varphi| \, d\varphi \, d\theta = \int_0^{2\pi} \int_0^{\pi/4} a^3 \cos \varphi \sin \varphi \, d\varphi \, d\theta \\ &= 2\pi a^3 \left(\frac{1}{4} \sin 2\varphi \Big|_0^{\pi/4} \right) = \frac{1}{2} \pi a^3.\end{aligned}$$

Therefore:

$$\bar{z} = \frac{1}{\text{area } S} \int_S z \, dS = \frac{\frac{1}{2} \pi a^3}{2\pi a^2(1 - 1/\sqrt{2})} = \frac{a}{4}(2 + \sqrt{2}).$$

4. Parametrization: $X(\theta, \varphi) = 2(\sin \varphi \cos \theta, \sin \varphi \sin \theta, \cos \varphi)$ where $z \geq 0$ and $0 \leq \varphi \leq \frac{\pi}{2}$. Since $x^2 + y^2 = 2x$, we have $4 \sin^2 \varphi = 4 \sin \varphi \cos \theta$, that is, $\sin \varphi = \cos \theta = \sin(\frac{\pi}{2} - \theta)$. But since $0 \leq \varphi \leq \frac{\pi}{2}$, $\varphi \leq \frac{\pi}{2} - \theta \leq \pi - \varphi$. Therefore $\varphi - \frac{\pi}{2} \leq \theta \leq \frac{\pi}{2} - \varphi$. Finally, $|X_\theta \times X_\varphi| = 4 \sin \varphi$.

Area:

$$\begin{aligned}\int_0^{\pi/2} \int_{\varphi-\pi/2}^{\pi/2-\varphi} |X_\theta \times X_\varphi| \, d\theta \, d\varphi &= \int_0^{\pi/2} \int_{\varphi-\pi/2}^{\pi/2-\varphi} 4 \sin \varphi \, d\theta \, d\varphi \\ &= \int_0^{\pi/2} (\pi - 2\varphi) 4 \sin \varphi \, d\varphi \\ &= 4\pi + 8[\varphi \cos \varphi - \sin \varphi]_0^{\pi/2} \\ &= 4\pi - 8.\end{aligned}$$

Centroid: By symmetry $\bar{y} = 0$.

$$\begin{aligned}\int_S x \, dS &= \int_0^{\pi/2} \int_{\varphi-\pi/2}^{\pi/2-\varphi} 8 \sin^2 \varphi \cos \theta \, d\theta \, d\varphi = \int_0^{\pi/2} 8 \sin^2 \varphi (2 \cos \varphi) \, d\varphi \\ &= \frac{16}{3} \sin^3 \varphi \Big|_0^{\pi/2} = \frac{16}{3}.\end{aligned}$$

Therefore:

$$\bar{x} = \frac{1}{\text{area } S} \int_S x \, dS = \frac{16/3}{4\pi - 8} = \frac{4}{3\pi - 6}.$$

Finally,

$$\begin{aligned}
 \int_S z \, dS &= \int_0^{\pi/2} \int_{\varphi-\pi/2}^{\pi/2-\varphi} 8 \cos \varphi \sin \varphi \, d\theta \, d\varphi \\
 &= \int_0^{\pi/2} (\pi - 2\varphi) 4 \sin 2\varphi \, d\varphi \\
 &= [-2\pi \cos 2\varphi + 4\varphi \cos 2\varphi - 2 \sin 2\varphi]_0^{\pi/2} \\
 &= 2\pi.
 \end{aligned}$$

Therefore

$$\bar{z} = \frac{1}{\text{area } S} \int_S z \, dS = \frac{2\pi}{4\pi - 8} = \frac{\pi}{2\pi - 4}.$$

□

Problem 16.3.3. For the following surfaces S with density function f , find the mass, average density, and center of mass.

1. $f(x, y, z) = xyz$, where S is the part of the plane $x + y + z = 1$ with $x, y, z > 0$.
2. $f(x, y, z) = x + y$, where S is the part of the plane $z = 2x + 3y$ inside the cylinder $x^2 + y^2 = 9$ with $x \geq 0, y \geq 0$.
3. $f(x, y, z) = z$, where S is given in cylindrical coordinates by $z = r^2$ inside the cylinder $r = 2$.
4. $f(x, y, z) = (x^2 + y^2)z$, where S is given in spherical coordinates by $\rho = 1$ with $z \geq 0$.
5. $f(x, y, z) = z^2$, where S is given by $x = \cos \theta, y = \sin \theta, z = z, 0 \leq \theta \leq 2\pi, 0 \leq z \leq 2\pi$.

Proof.

1. Parametrization: $X(x, y) = (x, y, 1 - x - y)$, where $x, y \geq 0, 1 - x - y \geq 0$.

$$X_x = (1, 0, -1), \quad X_y = (0, 1, -1), \quad |X_x \times X_y| = \sqrt{3}.$$

$$\text{Area: } \frac{\sqrt{3}}{2}.$$

Mass:

$$M = \iint_S f \, dS = \int_0^1 \int_0^{1-x} xy(1 - x - y) \sqrt{3} \, dy \, dx = \frac{\sqrt{3}}{120}.$$

$$\text{Average density: } \frac{M}{\text{area } S} = \frac{1}{60}.$$

Center of Mass: By symmetry, $\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$.

2. Parametrization: $X(r, \theta) = r(\cos \theta, \sin \theta, 2 \cos \theta + 3 \sin \theta)$ where $x, y \geq 0$, $x^2 + y^2 \leq 9$. $X_r = (\cos \theta, \sin \theta, 2 \cos \theta + 3 \sin \theta)$, $X_\theta = r(-\sin \theta, \cos \theta, -2 \sin \theta + 3 \cos \theta)$, and $|X_r \times X_\theta| = |r(-2, -3, 1)| = \sqrt{14}r$.

Area:

$$\int_0^{\pi/2} \int_0^3 \sqrt{14}r \, dr \, d\theta = \frac{9}{4} \sqrt{14} \pi.$$

Mass:

$$\int_0^{\pi/2} \int_0^3 (r \cos \theta + r \sin \theta) \sqrt{14}r \, dr \, d\theta = 18\sqrt{14}.$$

Average density: $\frac{M}{\text{area } S} = \frac{8}{\pi}.$

CoM:

$$\begin{aligned} \bar{x} &= \frac{1}{M} \int_0^{\pi/2} \int_0^3 r \cos \theta (r \cos \theta + r \sin \theta) \sqrt{14}r \, dr \, d\theta \\ &= \frac{1}{18\sqrt{14}} \cdot \frac{81}{4} \left(\frac{\pi}{4} + \frac{1}{2} \right) \sqrt{14} = \frac{9}{32}(\pi + 2), \\ \bar{y} &= \frac{1}{M} \int_0^{\pi/2} \int_0^3 r \sin \theta (r \cos \theta + r \sin \theta) \sqrt{14}r \, dr \, d\theta = \frac{9}{32}(\pi + 2), \\ \bar{z} &= \frac{1}{M} \iint_S (2x + 3y) \, dS = 2\bar{x} + 3\bar{y} = \frac{45}{32}(\pi + 2). \end{aligned}$$

3. Parametrization: $X(r, \theta) = (r \cos \theta, r \sin \theta, r^2)$, where $r \leq 2$. $X_r = (\cos \theta, \sin \theta, 2r)$, $X_\theta = (-r \sin \theta, r \cos \theta, 0)$, and $|X_r \times X_\theta| = |(-2r^2 \cos \theta, -2r^2 \sin \theta, r)| = r\sqrt{1 + 4r^2}$.

Area:

$$\iint_S dS = \int_0^{2\pi} \int_0^2 r\sqrt{1 + 4r^2} \, dr \, d\theta = 2\pi \cdot \frac{1}{12} (1 + 4r^2)^{3/2} \Big|_0^2 = \frac{\pi}{6} (17\sqrt{17} - 1).$$

Mass:

$$\int_0^{2\pi} \int_0^2 r^3 \sqrt{1 + 4r^2} \, dr \, d\theta = \frac{\pi}{60} (391\sqrt{17} + 1).$$

Average density: $\frac{M}{\text{area } S} = \frac{391\sqrt{17} + 1}{10(17\sqrt{17} - 1)}.$

CoM: By symmetry $\bar{x} = \bar{y} = 0$.

$$\bar{z} = \frac{1}{M} \int_0^{2\pi} \int_0^2 r^5 \sqrt{1 + 4r^2} \, dr \, d\theta = \frac{7769\sqrt{17} - 1}{7(391\sqrt{17} + 1)}.$$

4. Parametrization: $X(\theta, \varphi) = (\sin \varphi \cos \theta, \sin \varphi \sin \theta, \cos \varphi)$, where $0 \leq \varphi \leq \frac{\pi}{2}$, $0 \leq \theta \leq 2\pi$. $|X_\theta \times X_\varphi| = \sin \varphi$.

Area: 2π .

Mass:

$$\int_0^{2\pi} \int_0^{\pi/2} \sin^2 \varphi \cos \varphi \sin \varphi \, d\varphi \, d\theta = 2\pi \cdot \frac{1}{4} \sin^4 \varphi \Big|_0^{\pi/2} = \frac{\pi}{2}.$$

Average density: $\frac{M}{\text{area } S} = \frac{1}{4}$.

CoM: By symmetry $\bar{x} = \bar{y} = 0$.

$$\begin{aligned} \bar{z} &= \frac{1}{M} \int_0^{2\pi} \int_0^{\pi/2} \cos \varphi \sin^2 \varphi \cos \varphi \sin \varphi \, d\varphi \, d\theta \\ &= \frac{2}{\pi} \cdot 2\pi \left(-\frac{1}{3} \cos^3 \varphi + \frac{1}{5} \cos^5 \varphi \Big|_0^{\pi/2} \right) = \frac{8}{15}. \end{aligned}$$

5. Parametrization: $X(\theta, z) = (\cos \theta, \sin \theta, z)$, where $0 \leq \theta \leq 2\pi$, $0 \leq z \leq 2\pi$.

$X_\theta = (-\sin \theta, \cos \theta, 0)$, $X_z = (0, 0, 1)$, and $|X_\theta \times X_z| = 1$.

Area: $4\pi^2$.

Mass:

$$\int_0^{2\pi} \int_0^{2\pi} z^2 \, d\theta \, dz = \frac{16}{3} \pi^4.$$

Average density: $\frac{M}{\text{area } S} = \frac{4}{3} \pi^2$.

CoM: By symmetry $\bar{x} = \bar{y} = 0$.

$$\bar{z} = \frac{1}{M} \int_0^{2\pi} \int_0^{2\pi} z \cdot z^2 \, d\theta \, dz = \frac{3}{16\pi^4} \cdot 2\pi \cdot \frac{(2\pi)^4}{4} = \frac{3}{2} \pi.$$

□

Problem 16.4.1. Compute $\iint_S \mathbf{F} \cdot d\mathbf{S}$.

1. $\vec{F} = x\vec{i} + y\vec{j} + z\vec{k}$, where S is the portion of the plane $2x + 2y + z = 3$ with $x, y, z > 0$.
2. $\vec{F} = 2y\vec{i} + 3z\vec{k}$, where S is the portion of the plane $z = 3x + 2$ inside the cylinder $x^2 + y^2 = 4$.
3. $\vec{F} = z\vec{k}$, where S is the upper hemisphere of the sphere represented in spherical coordinates by $\rho = 2$.
4. $\vec{F} = y\vec{i} - x\vec{j}$, where S is the surface represented in cylindrical coordinates by $z = r$, inside the cylinder $r = 3$.
5. $\vec{F} = 2x\vec{i} + 2y\vec{j} + 3\vec{k}$, where S is the portion of the paraboloid $z = 4 - x^2 - y^2$ that lies above the xy -plane.
6. $\vec{F} = 3xy^2\vec{i} + 3x^2y\vec{j} + z^3\vec{k}$, where S is the sphere $x^2 + y^2 + z^2 = 1$.

Proof.

1. Parametrization: $X(x, y) = (x, y, 3 - 2x - 2y)$, $x, y \geq 0$, $3 - 2x - 2y \geq 0$.

$$X_x = (1, 0, -2), X_y = (0, 1, -2), X_x \times X_y = (2, 2, 1).$$

$$\begin{aligned} \iint_S \mathbf{F} \cdot d\vec{S} &= \int_0^{3/2} \int_0^{3/2-y} (x, y, 3 - 2x - 2y) \cdot (2, 2, 1) dx dy \\ &= \int_0^{3/2} \int_0^{3/2-x} 3 dx dy = \frac{27}{8}. \end{aligned}$$

2. Parametrization: $X(r, \theta) = (r \cos \theta, r \sin \theta, 3r \cos \theta + 2)$, $r \leq 2$, $0 \leq \theta \leq 2\pi$.

$$X_r = (\cos \theta, \sin \theta, 3 \cos \theta), X_\theta = r(-\sin \theta, \cos \theta, -3 \sin \theta),$$

$$X_r \times X_\theta = r(-3, 0, 1).$$

$$\begin{aligned} \iint_S \mathbf{F} \cdot d\vec{S} &= \int_0^{2\pi} \int_0^2 (2r \sin \theta, 0, 3(3r \cos \theta + 2)) \cdot r(-3, 0, 1) dr d\theta \\ &= \int_0^{2\pi} \int_0^2 (9r \cos \theta - 6r \sin \theta + 6)r dr d\theta = 24\pi. \end{aligned}$$

3. Parametrization: $X(\phi, \theta) = 2(\sin \phi \cos \theta, \sin \phi \sin \theta, \cos \phi)$, $0 \leq \phi \leq \frac{\pi}{2}$, $0 \leq \theta \leq 2\pi$.

$$X_\phi = 2(\cos \phi \cos \theta, \cos \phi \sin \theta, -\sin \phi), X_\theta = 2(-\sin \phi \sin \theta, \sin \phi \cos \theta, 0).$$

$$X_\phi \times X_\theta = 4 \sin \phi (\sin \phi \cos \theta, \sin \phi \sin \theta, \cos \phi).$$

$$\begin{aligned} \iint_S \vec{F} \cdot d\vec{S} &= \int_0^{2\pi} \int_0^{\pi/2} 2(0, 0, \cos \phi) \cdot 4 \sin \phi (\sin \phi \cos \theta, \sin \phi \sin \theta, \cos \phi) d\phi d\theta = \\ &= \int_0^{2\pi} \int_0^{\pi/2} 8 \cos^2 \phi \sin \phi d\phi d\theta = \frac{16}{3}\pi. \end{aligned}$$

4. Parametrization: $X(r, \theta) = (r \cos \theta, r \sin \theta, r)$, $r \leq 3$, $0 \leq \theta \leq 2\pi$.

$$X_r = (\cos \theta, \sin \theta, 1), \quad X_\theta = r(\sin \theta, -\cos \theta, 0), \quad X_r \times X_\theta = r(-\cos \theta, -\sin \theta, 1).$$

$$\begin{aligned} \iint_S \vec{F} \cdot d\vec{S} &= \int_0^{2\pi} \int_0^3 (r \sin \theta, -r \cos \theta, 0) \cdot r(-\cos \theta, -\sin \theta, 1) dr d\theta = \\ &= \int_0^{2\pi} \int_0^3 0 dr d\theta = 0. \end{aligned}$$

5. Parametrization: $X(r, \theta) = (r \cos \theta, r \sin \theta, 4 - r^2)$, $r \leq 2$, $0 \leq \theta \leq 2\pi$.

$$X_r = (\cos \theta, \sin \theta, -2r), \quad X_\theta = r(-\sin \theta, \cos \theta, 0), \quad X_r \times X_\theta = r(2r \cos \theta, 2r \sin \theta, 1).$$

$$\begin{aligned} \iint_S \vec{F} \cdot d\vec{S} &= \int_0^{2\pi} \int_0^2 (2r \cos \theta, 2r \sin \theta, 3) \cdot r(2r \cos \theta, 2r \sin \theta, 1) dr d\theta = \\ &= \int_0^{2\pi} \int_0^2 (4r^2 + 3)r dr d\theta = 44\pi. \end{aligned}$$

6. Parametrization: $X(\phi, \theta) = (\sin \phi \cos \theta, \sin \phi \sin \theta, \cos \phi)$, $0 \leq \phi \leq \pi$, $0 \leq \theta \leq 2\pi$.

$$X_\phi = (\cos \phi \cos \theta, \cos \phi \sin \theta, -\sin \phi), \quad X_\theta = (-\sin \phi \sin \theta, \sin \phi \cos \theta, 0).$$

$$X_\phi \times X_\theta = \sin \phi (\sin \phi \cos \theta, \sin \phi \sin \theta, \cos \phi).$$

$$\begin{aligned} \iint_S \vec{F} \cdot d\vec{S} &= \iint_S (3xy^2, 3x^2y, z^3) \cdot (x, y, z) dS = \iint_S (6x^2y^2 + z^4) dS. \\ &= \int_0^{2\pi} \int_0^\pi (6 \sin^4 \phi \cos^2 \theta \sin^2 \theta + \cos^4 \phi) \sin \phi d\phi d\theta = \frac{12}{5}\pi. \end{aligned}$$

□

Problem 16.4.3. Let S be the boundary of the region enclosed by the paraboloid $z = 4 - x^2 - y^2$ and the xy -plane. For the vector field $\mathbf{F} = 2x\mathbf{i} + 2y\mathbf{j} + 3z\mathbf{k}$, find the flux leaving the surface S .

Proof.

Let the upper surface of S be S_1 and the lower surface be S_2 . For S_1 , we may use the result of the (5) of the previous problem;

$$\begin{aligned} \iint_{S_1} \mathbf{F} \cdot d\mathbf{S} &= \int_0^{2\pi} \int_0^2 (2r \cos \theta, 2r \sin \theta, 3) \cdot r(2r \cos \theta, 2r \sin \theta, 1) dr d\theta \\ &= \int_0^{2\pi} \int_0^2 (4r^2 \cos^2 \theta + 4r^2 \sin^2 \theta + 3)r dr d\theta = 28\pi \end{aligned}$$

For S_2 , it is clear that $\mathbf{n} = (0, 0, -1)$, and using this we compute

$$\iint_{S_2} \mathbf{F} \cdot d\mathbf{S} = \iint_{S_2} (2x, 2y, 3) \cdot (0, 0, -1) dS = (-3) \text{ area } S_2 = -12\pi$$

Therefore, the answer is $28\pi - 12\pi = 16\pi$.

□

Problem 16.4.4. Let S be the boundary of the region enclosed by the paraboloids $z = x^2 + y^2$ and $z = 18 - x^2 - y^2$. Find the flux of the vector field $\mathbf{F} = z^2\mathbf{k}$ leaving S .

Proof. The intersection curve of the two surfaces is given by $z = 9, x^2 + y^2 = 9$, and S consists of the following two surfaces:

$$S_1 : z = x^2 + y^2, \quad z \leq 9,$$

$$S_2 : z = 18 - (x^2 + y^2), \quad z \geq 9.$$

Parametrize S_1 and S_2 by

$$X(r, \theta) = (r \cos \theta, -r \sin \theta, r^2), \quad r \leq 3, 0 \leq \theta \leq 2\pi,$$

$$Y(r, \theta) = (r \cos \theta, r \sin \theta, 18 - r^2), \quad r \leq 3, 0 \leq \theta \leq 2\pi.$$

With similar computations, we obtain

$$X_r \times X_\theta = r(2r \cos \theta, 2r \sin \theta, -1), \quad Y_r \times Y_\theta = r(2r \cos \theta, 2r \sin \theta, 1).$$

Now, computing the flux leaving S_1 and S_2 separately, we have

$$\begin{aligned} \iint_{S_1} \mathbf{F} \cdot d\mathbf{S} &= \int_0^{2\pi} \int_0^3 (0, 0, (r^2)^2) \cdot r(2r \cos \theta, 2r \sin \theta, -1) dr d\theta \\ &= \int_0^{2\pi} \int_0^3 -r^5 dr d\theta \\ &= -243\pi, \end{aligned}$$

$$\begin{aligned} \iint_{S_2} \mathbf{F} \cdot d\mathbf{S} &= \int_0^{2\pi} \int_0^3 (0, 0, (18 - r^2)^2) \cdot r(2r \cos \theta, 2r \sin \theta, 1) dr d\theta \\ &= \int_0^{2\pi} \int_0^3 (18 - r^2)^2 r dr d\theta \\ &= 1701\pi. \end{aligned}$$

Therefore,

$$\begin{aligned} \iint_{S_1} \mathbf{F} \cdot d\mathbf{S} + \iint_{S_2} \mathbf{F} \cdot d\mathbf{S} &= -243\pi + 1701\pi \\ &= 1458\pi. \end{aligned}$$

□

Problem 16.4.5. For the normal vector field $\mathbf{N}(u, v)$ obtained from the regular surface $X(u, v)$ and the normal vector field $\tilde{\mathbf{N}}(\tilde{u}, \tilde{v})$ obtained from an orientation-preserving reparametrization $\tilde{X}(\tilde{u}, \tilde{v})$ of this surface, show that

$$\frac{\mathbf{N}}{|\mathbf{N}|} = \frac{\tilde{\mathbf{N}}}{|\tilde{\mathbf{N}}|}.$$

Proof. Let G be an orientation preserving reparametrization such that $\tilde{X} = X \circ G$.

In particular, G is an invertible C^1 map. Then by the chain rule we get:

$$\begin{aligned}\tilde{X}_{\tilde{u}} &= X_u \frac{\partial u}{\partial \tilde{u}} + X_v \frac{\partial v}{\partial \tilde{u}} \\ \tilde{X}_{\tilde{v}} &= X_u \frac{\partial u}{\partial \tilde{v}} + X_v \frac{\partial v}{\partial \tilde{v}}.\end{aligned}$$

Therefore,

$$\begin{aligned}\tilde{\mathbf{N}} &= \tilde{X}_{\tilde{u}} \times \tilde{X}_{\tilde{v}} \\ &= \left(\frac{\partial u}{\partial \tilde{u}} X_u + \frac{\partial v}{\partial \tilde{u}} X_v \right) \times \left(\frac{\partial u}{\partial \tilde{v}} X_u + \frac{\partial v}{\partial \tilde{v}} X_v \right) \\ &= \left(\frac{\partial u}{\partial \tilde{u}} \frac{\partial v}{\partial \tilde{v}} - \frac{\partial u}{\partial \tilde{v}} \frac{\partial v}{\partial \tilde{u}} \right) (X_u \times X_v) \\ &= \det(dG)(\mathbf{N}).\end{aligned}$$

Since $\det G > 0$, we are done. □

6

Divergence Theorem

6.1 problem

Problem 17.1.1. Show that the following cases are compatible with divergence theorem.

1. $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, $S : x^2 + y^2 + z^2 = 1$
2. $\mathbf{F} = |\mathbf{r}|\mathbf{r}$, $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, $S : x^2 + y^2 + z^2 = 9$
3. $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, S is the boundary of the region enclosed by the three coordinate planes and the planes $x = 2, y = 2, z = 2$.
4. $\mathbf{F} = xy\mathbf{i} + yz\mathbf{j} + zx\mathbf{k}$, S is the surface in problem (3).
5. $\mathbf{F} = (x + y)\mathbf{i} + (y + z)\mathbf{j} + (z + x)\mathbf{k}$, S is the boundary of the tetrahedron enclosed by the three coordinate planes and the plane $x + y + z = 1$.

Proof.

1. Since $\mathbf{n} = (x, y, z)$,

$$\begin{aligned} \iint_S \mathbf{F} \cdot d\mathbf{S} &= \iint_S (x, y, z) \cdot (x, y, z) dA = \iint_S 1 dA = 4\pi, \\ \iiint_{\text{int } S} \text{div } \mathbf{F} dV &= \iiint_{\text{int } S} 3 dV = 3 \cdot \frac{4}{3}\pi 1^3 = 4\pi. \end{aligned}$$

2. Since $\mathbf{n} = \frac{1}{3}(x, y, z)$ on S , $\mathbf{F} = (x, y, z) = 3(x, y, z)$. On the other hand, inside

the surface, $\operatorname{div} \mathbf{F} = 4\sqrt{x^2 + y^2 + z^2}$. Therefore,

$$\begin{aligned}\iint_S \mathbf{F} \cdot d\mathbf{S} &= \iint_S 3(x, y, z) \cdot \frac{1}{3}(x, y, z) dA = \iint_S 9 dA = 9 \cdot 4\pi 3^2 = 324\pi, \\ \iiint_{\text{int } S} \operatorname{div} \mathbf{F} dV &= \int_0^{2\pi} \int_0^\pi \int_0^3 4\rho \cdot \rho^2 \sin \phi d\rho d\phi d\theta = 324\pi.\end{aligned}$$

One should make clear that $\mathbf{F}$ is indeed C^1 by the definition.

3. By symmetry, it suffices to compute only the face parallel to the xy -plane and then multiply the result by 3. Let $S_1 = S \cap \{(x, y, z) : z = 2\}$, $S_2 = S \cap \{(x, y, z) : z = 0\}$. The normal vectors of S_1 and S_2 are both $\mathbf{n} = (0, 0, -1)$. Therefore,

$$\begin{aligned}\iint_S \mathbf{F} \cdot d\mathbf{S} &= 3 \left(\iint_{S_1} \mathbf{F} \cdot d\mathbf{S} + \iint_{S_2} \mathbf{F} \cdot d\mathbf{S} \right) \\ &= 3 \left(\iint_{S_1} 2 dA + \iint_{S_2} 0 dA \right) = 3 \cdot (2 \cdot \operatorname{Area}(S_1) + 0) = 24.\end{aligned}$$

On the other hand,

$$\iiint_{\text{int } S} \operatorname{div} \mathbf{F} dV = \iiint_{\text{int } S} 3\mathbf{F} dV = 3 \cdot 8 = 24.$$

4. Using the same S_1, S_2 as in (3),

$$\begin{aligned}\iint_{S_1} \mathbf{F} \cdot d\mathbf{S} + \iint_{S_2} \mathbf{F} \cdot d\mathbf{S} &= \iint_{S_1} (xy, 2y, 2x) \cdot (0, 0, 1) dA + \iint_{S_2} (xy, 0, 0) \cdot (0, 0, -1) dA \\ &= \iint_{S_1} 2x dA + \iint_{S_2} 0 dA = 2 \operatorname{Area}(S_1) \cdot \bar{x} + 0 = 2 \cdot 4 \cdot 1 = 8.\end{aligned}$$

By symmetry, $\iint_S \mathbf{F} \cdot d\mathbf{S} = 3 \cdot 8 = 24$. On the other hand,

$$\iiint_{\text{int } S} \operatorname{div} \mathbf{F} dV = \int_0^2 \int_0^2 \int_0^2 (y + z + x) dx dy dz = 24.$$

5. For $S_1 : S \cap \{(x, y, z) : x + y + z = 1\}$, we have $\mathbf{n} = \frac{1}{\sqrt{3}}(1, 1, 1)$, so

$$\iint_{S_1} \mathbf{F} \cdot d\mathbf{S} = \iint_{S_1} (x + y, y + z, x + z) \cdot \frac{1}{\sqrt{3}}(1, 1, 1) dA = \iint_{S_1} \frac{2}{\sqrt{3}} dA = 1.$$

The remaining faces have the same value by symmetry, so we compute only

$$S_2 : S \cap \{(x, y, z) : z = 0\}.$$

$$\begin{aligned} \iint_{S_2} \mathbf{F} \cdot d\mathbf{S} &= \iint_{S_2} (x + y, y + z, x + z) \cdot (0, 0, -1) dA = \iint_{S_2} (-x) dA \\ &= (-1) \text{Area}(S_2) \cdot \bar{x} = -\frac{1}{2} \cdot \frac{1}{3} = -\frac{1}{6}. \end{aligned}$$

Therefore,

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = 1 + 3 \cdot \left(-\frac{1}{6}\right) = \frac{1}{2}.$$

On the other hand,

$$\iiint_{\text{int } S} \text{div } \mathbf{F} dV = \iiint_{\text{int } S} 3 dV = 3 \cdot \frac{1}{6} = \frac{1}{2}.$$

□

Problem 17.1.2. For the vector field $\mathbf{F} = 3xy^2\mathbf{i} + 3x^2y\mathbf{j} + z^3\mathbf{k}$, find the flux leaving the standard unit sphere $S : x^2 + y^2 + z^2 = 1$.

Proof. By the divergence theorem,

$$\begin{aligned} \iint_S \mathbf{F} \cdot d\mathbf{S} &= \iiint_{\text{int } S} \text{div } \mathbf{F} dV = \iiint_{\text{int } S} 3(y^2 + x^2 + z^2) dV \\ &= \int_0^{2\pi} \int_0^\pi \int_0^1 3\rho^2 \rho^2 \sin \phi d\rho d\phi d\theta \\ &= 3 \cdot \int_0^{2\pi} d\theta \cdot \int_0^\pi \sin \phi d\phi \cdot \int_0^1 \rho^4 d\rho = 3 \cdot 2\pi \cdot 2 \cdot \frac{1}{5} \\ &= \frac{12}{5}\pi. \end{aligned}$$

□

Problem 17.1.3. For the outward unit normal vector field $\mathbf{n} = n_1\mathbf{i} + n_2\mathbf{j} + n_3\mathbf{k}$ of a closed surface S , show that the volume V of the region inside S satisfies

$$V = \iint_S n_1 x dS = \iint_S n_2 y dS = \iint_S n_3 z dS.$$

Proof. Without loss of generality, enough to show only the first identity.

$$\begin{aligned}
 \iint_S n_1 x \, dS &= \iint_S (x, 0, 0) \cdot \mathbf{n} \, dS \\
 &= \iint_S (x, 0, 0) \cdot d\mathbf{S} \\
 &= \iiint_{\text{int } S} \text{div}(x, 0, 0) \, dV \\
 &= \iiint_{\text{int } S} 1 \, dV \\
 &= V.
 \end{aligned}$$

□

Problem 17.1.4. Let $\mathbf{F}$ be a C^1 vector field defined on an open region U in space. If for every ball B contained in U ,

$$\iint_{\partial B} \mathbf{F} \cdot d\mathbf{S} = 0,$$

show that $\mathbf{F}$ is incompressible.

Proof. We show that for all $p \in U$ we have $\text{div } F(p) = 0$. To derive a contradiction, suppose that there exists $p \in U$ such that $\text{div } F(p) \neq 0$.

WLOG WMA $\text{div } f(p) = \epsilon > 0$. Since $\mathbf{F}$ is C^1 , $\text{div } F$ is continuous. Therefore, by the definition of continuity, there exists $\delta > 0$ such that

$$|X - p| < \delta \Rightarrow |\text{div } F(X) - \text{div } F(p)| < \frac{\epsilon}{2}.$$

That is, if $X \in B_\delta(p)$, then $\frac{\epsilon}{2} < \text{div } F(X) < \frac{3\epsilon}{2}$. But then:

$$\begin{aligned}
 \iiint_{B_\delta(p)} \text{div } \mathbf{F} \, dV &> \iiint_{B_\delta(p)} \frac{\epsilon}{2} \, dV \\
 &= \frac{\epsilon}{2} \text{Vol}(B_\delta(p)) > 0.
 \end{aligned}$$

This is a contradiction since by the divergence theorem,

$$0 < \iiint_{B_\delta(p)} \text{div } \mathbf{F} \, dV = \iint_{\partial B_\delta(p)} \mathbf{F} \cdot d\mathbf{S} = 0.$$

□

Problem 17.2.1. For the vector field $A = \frac{\mathbf{r}}{|\mathbf{r}|^3}$, find the amount of flux leaving each of the following surfaces. The unit normal vector n determining the orientation is chosen so that $n \cdot k \geq 0$.

1. The part of the sphere $x^2 + y^2 + z^2 = 4$ where $z \geq 0$.
2. The part of the sphere $x^2 + y^2 + z^2 = 9$ where $z \leq 0$.
3. The part of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ where x, y, z are all nonnegative.

Proof. Since integration of A is exactly the area of projected surface onto sphere, we get:

1. 2π
2. -2π
3. $\frac{\pi}{2}$

□

7

Curl and Stokes Theorem

7.1 problem

Problem 18.1.1. Find the curl of the following vector fields.

1. $\mathbf{F}(x, y, z) = y\mathbf{i} + x\mathbf{j}$

2. $\mathbf{F}(x, y, z) = (x^2 + y^2 + z^2)(3, 4, 5)$

Proof. 1. $\operatorname{curl} \mathbf{F} = (0 - D_3x, 0, D_1x - D_2y) = (0, 0, 0)$

2.

$$\begin{aligned}\operatorname{curl} \mathbf{F} &= (D_25y^2 - D_34z^2, D_33z^2 - D_15x^2, D_14x^2 - D_23x^2) \\ &= 2(5y - 4z, 3z - 5x, 4x - 3y)\end{aligned}$$

□

Problem 18.1.4. For vector fields $\mathbf{F}$ and $\mathbf{G}$, show that

$$\operatorname{div}(\mathbf{F} \times \mathbf{G}) = \operatorname{curl} \mathbf{F} \cdot \mathbf{G} - \mathbf{F} \cdot \operatorname{curl} \mathbf{G}.$$

Proof. Let $\mathbf{F} = (f_1, f_2, f_3)$ and $\mathbf{G} = (g_1, g_2, g_3)$. Then

$$\begin{aligned} \operatorname{div}(\mathbf{F} \times \mathbf{G}) &= \operatorname{div}(f_2g_3 - f_3g_2, f_3g_1 - f_1g_3, f_1g_2 - f_2g_1) \\ &= D_1(f_2g_3 - f_3g_2) + D_2(f_3g_1 - f_1g_3) + D_3(f_1g_2 - f_2g_1) \\ &= (D_1f_2)g_3 + f_2(D_1g_3) - (D_1f_3)g_2 - f_3(D_1g_2) \\ &\quad + (D_2f_3)g_1 + f_3(D_2g_1) - (D_2f_1)g_3 - f_1(D_2g_3) \\ &\quad + D_3(f_1)g_2 + f_1(D_3g_2) - (D_3f_2)g_1 - f_2(D_3g_1) \\ &= (D_2f_3 - D_3f_2)g_1 + (D_3f_1 - D_1f_3)g_2 + (D_1f_2 - D_2f_1)g_3 \\ &\quad - [f_1(D_2g_3 - D_3g_2) + f_2(D_3g_1 - D_1g_3) + f_3(D_1g_2 - D_2g_1)] \\ &= \operatorname{curl} \mathbf{F} \cdot \mathbf{G} - \mathbf{F} \cdot \operatorname{curl} \mathbf{G}. \end{aligned}$$

□

Problem 18.2.1. For the upper-sphere $z = \sqrt{1 - x^2 - y^2}$ and the vector field $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, show that Stokes' theorem holds.

Proof.

$$1. \operatorname{curl} \mathbf{F} = (D_2z - D_3y, D_3x - D_1z, D_1y - D_2z) = 0,$$

$$\therefore \iint_S \operatorname{curl} \mathbf{F} \cdot d\mathbf{S} = 0$$

$$2. \int_{\partial S} \mathbf{F} \cdot d\mathbf{s} = 0$$

Let us parametrize the unit circle in the xy -plane, which is the boundary of S , by $X(t) = (\cos t, \sin t, 0)$, $0 \leq t \leq 2\pi$. If we choose the orientation of S so that $\mathbf{n} \cdot \mathbf{k} \geq 0$, then the orientation of $X(t)$ agrees with that of ∂S .

$$\int_{\partial S} \mathbf{F} \cdot d\mathbf{s} = \int_0^{2\pi} (\cos t, \sin t, 0) \cdot (-\sin t, \cos t, 0) dt = 0$$

Therefore the Stokes' theorem holds.

□

Problem 18.2.4. Let

$$S_1 : x^2 + y^2 = 1, \quad 0 \leq z \leq 1$$

and

$$S_2 : x^2 + y^2 + (z - 1)^2 = 1, \quad z \geq 1,$$

and let $S = S_1 \cup S_2$. For

$$\mathbf{F} = (xz + yz^2 + x)\mathbf{i} + (xyz^3 + y)\mathbf{j} + x^2z^4\mathbf{k},$$

compute

$$\iint_S \operatorname{curl} \mathbf{F} \cdot d\mathbf{S}.$$

(The orientation of S is chosen so that $\mathbf{n} \cdot \mathbf{k} \geq 0$.)

Proof. By Stokes' theorem,

$$\iint_S \operatorname{curl} \mathbf{F} \cdot d\mathbf{S} = \oint_{\partial S} \mathbf{F} \cdot d\mathbf{s}.$$

It is clear that ∂S is the unit circle in the xy -plane. Parametrize the boundary by

$$X(t) = (\cos t, \sin t, 0), \quad 0 \leq t \leq 2\pi.$$

If we choose the orientation of S so that $\mathbf{n} \cdot \mathbf{k} \geq 0$, then the orientation of $X(t)$ agrees with that of ∂S . Now, computing the line integral on the right-hand side,

$$\oint_{\partial S} \mathbf{F} \cdot d\mathbf{s} = \int_0^{2\pi} \mathbf{F}(X(t)) \cdot X'(t) dt = \int_0^{2\pi} (\cos t, \sin t, 0) \cdot (-\sin t, \cos t, 0) dt = 0.$$

Therefore the answer is 0. □

Problem 18.2.5. Define a surface

$$S : 2z = x^2 + y^2, \quad z \leq 2$$

and vector field

$$\mathbf{F} = 3y\mathbf{i} - xy\mathbf{j} + yz^2\mathbf{k}.$$

Compute $\iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S}$. (The orientation of S is chosen so that $\mathbf{n} \cdot \mathbf{k} > 0$.)

Proof. By Stokes' theorem,

$$\iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S} = \int_{\partial S} \mathbf{F} \cdot d\mathbf{s}.$$

Clearly we see that ∂S is the circle of radius 2 in the plane $z = 2$ centered at $(0, 0, 2)$. Parametrize it by

$$X(t) = (2 \cos t, 2 \sin t, 2), \quad 0 \leq t \leq 2\pi,$$

and compute the line integral on the right-hand side:

$$\begin{aligned} \int_{\partial S} \mathbf{F} \cdot d\mathbf{s} &= \int_0^{2\pi} \mathbf{F}(X(t)) \cdot X'(t) dt \\ &= \int_0^{2\pi} (6 \sin t, -4 \cos t \sin t, 8 \sin t) \cdot (-2 \sin t, 2 \cos t, 0) dt \\ &= \int_0^{2\pi} (-12 \sin^2 t - 8 \sin t \cos^2 t) dt \\ &= -12\pi. \end{aligned}$$

□

A

Appendix

A.1 Vector Spaces

In order to discuss about tangent space, we need to study the vector space and its basis. Before that, we first define the vector space. Take $\mathbb{R}^2$ as a spirit of this definition.

Reference: Friedberg Linear Algebra. Almost every contents are same with contents of it, I just collected what we will need in this course.

Definition A.1. A *vector space* (or *linear space*) V over a field² F consists of a set on which two operations (called **addition** and **scalar multiplication**, respectively) are defined so that for each pair of elements x, y , in V there is a unique element $x + y$ in V , and for each element a in F and each element x in V there is a unique element ax in V , such that the following conditions hold.

(VS 1) For all x, y in V , $x + y = y + x$ (commutativity of addition).

(VS 2) For all x, y, z in V , $(x + y) + z = x + (y + z)$ (associativity of addition).

(VS 3) There exists an element in V denoted by 0 such that $x + 0 = x$ for each x in V .

(VS 4) For each element x in V there exists an element y in V such that $x + y = 0$.

(VS 5) For each element x in V , $1x = x$.

(VS 6) For each pair of elements a, b in F and each element x in V , $(ab)x = a(bx)$.

(VS 7) For each element a in F and each pair of elements x, y in V , $a(x + y) = ax + ay$.

(VS 8) For each pair of elements a, b in F and each element x in V , $(a + b)x = ax + bx$.

The elements $x + y$ and ax are called the **sum** of x and y and the **product** of a and x , respectively.

Example A.1.1. The set of all n -tuples with entries from a field F is denoted by F^n .

This set is a vector space over F with the operations of coordinatewise addition and scalar multiplication; that is, if $u = (a_1, a_2, \dots, a_n) \in F^n$, $v = (b_1, b_2, \dots, b_n) \in F^n$, and $c \in F$, then

$$u + v = (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n) \quad \text{and} \quad cu = (ca_1, ca_2, \dots, ca_n).$$

Thus $\mathbb{R}^3$ is a vector space over $\mathbb{R}$. In this vector space,

$$(3, -2, 0) + (-1, 1, 4) = (2, -1, 4) \quad \text{and} \quad -5(1, -2, 0) = (-5, 10, 0).$$

Similarly, $\mathbb{C}^2$ is a vector space over $\mathbb{C}$. In this vector space,

$$(1 + i, 2) + (2 - 3i, 4i) = (3 - 2i, 2 + 4i) \quad \text{and} \quad i(1 + i, 2) = (-1 + i, 2i).$$

Vectors in F^n may be written as **column vectors**

$$\begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}$$

rather than as **row vectors** $(a_1, a_2, \dots, a_n)$. Since a 1-tuple whose only entry is from F can be regarded as an element of F , we usually write F rather than F^1 for the vector space of 1-tuples with entry from F . ♦

Now, instead of writing all the elements of a vector space, what we want to do is to represent a vector space with some of elements of them, as we did in topology. Doing such thing is a **basis** of a vector space. We are going to prepare to define it.

Definition A.2. Let S be a nonempty subset of a vector space V . The **span** of S , denoted $\text{span}(S)$, is the set consisting of all linear combinations of the vectors in S . For convenience, we define $\text{span}(\emptyset) = \{0\}$.

In $\mathbb{R}^3$, for instance, the span of the set $\{(1, 0, 0), (0, 1, 0)\}$ consists of all vectors in $\mathbb{R}^3$ that have the form $a(1, 0, 0) + b(0, 1, 0) = (a, b, 0)$ for some scalars a and b . Thus the span of $\{(1, 0, 0), (0, 1, 0)\}$ contains all the points in the xy -plane. In this case, the span of the set is a subspace of $\mathbb{R}^3$. This fact is true in general.

As one can see, the span of a set “generates” something. We want to use this as a tool of generator. To do so, we follow the next flow:

Definition A.3. A subset W of a vector space V over a field F is called a **subspace** of V if W is a vector space over F with the operations of addition and scalar multiplication defined on V . In this case, we write $W \leq V$, instead of $W \subseteq V$.

That is, a subspace is a subset of a vector space which is also a vector space with the same operations. Luckily, we may only check the axioms (VS 3), (VS 4) since the others are already satisfied by the original space. Thus a subset W of a vector space V is a subspace if and only if the following four properties hold:

1. $x + y \in W$ whenever $x \in W$ and $y \in W$. (W is **closed under addition**.)
2. $cx \in W$ whenever $c \in F$ and $x \in W$. (W is **closed under scalar multiplication**.)
3. W has a zero vector.
4. Each vector in W has an additive inverse in W .¹

¹ In fact, this condition is not necessary.

Now, there is a good property of the span:

Theorem A.1. The span of any subset S of a vector space V is a subspace of V . Moreover, any subspace of V that contains S must also contain the span of S .

Proof. Just check the axioms.²

□

² or you may check the theorem 1.5 of the Friedberg Linear Algebra.

That is, in fact, the span “generates” a vector space.

Definition A.4. A subset S of a vector space V *generates* (or *spans*) V if $\text{span}(S) = V$. In this case, we also say that the vectors of S *generate* (or *span*) V .

Example A.1.2. The vectors $(1, 1, 0)$, $(1, 0, 1)$, and $(0, 1, 1)$ generate $\mathbb{R}^3$ since an arbitrary vector (a_1, a_2, a_3) in $\mathbb{R}^3$ is a linear combination of the three given vectors; in fact, the scalars r, s, t for which

$$r(1, 1, 0) + s(1, 0, 1) + t(0, 1, 1) = (a_1, a_2, a_3)$$

are

$$r = \frac{1}{2}(a_1 + a_2 - a_3), \quad s = \frac{1}{2}(a_1 - a_2 + a_3), \quad t = \frac{1}{2}(-a_1 + a_2 + a_3). \quad \blacklozenge$$

Example A.1.3. The polynomials $x^2 + 3x - 2$, $2x^2 + 5x - 3$, and $-x^2 - 4x + 4$ generate $P_2(\mathbb{R})$ since each of the three given polynomials belongs to $P_2(\mathbb{R})$ and each polynomial $ax^2 + bx + c$ in $P_2(\mathbb{R})$ is a linear combination of these three, namely,

$$\begin{aligned} &(-8a + 5b + 3c)(x^2 + 3x - 2) + (4a - 2b - c)(2x^2 + 5x - 3) \\ &+ (-a + b + c)(-x^2 - 4x + 4) = ax^2 + bx + c. \quad \blacklozenge \end{aligned}$$

Now we can define the basis of a vector space.

Definition A.5. A *basis* β for a vector space V is a linearly independent subset of V that generates V . If β is a basis for V , we also say that the vectors of β form a basis for V .

Indeed, one may observe that the basis behaves like the **axis** of the Euclidean space. Note that the basis may not be unique.

Example A.1.4. In F^n , let $e_1 = (1, 0, 0, \dots, 0)$, $e_2 = (0, 1, 0, \dots, 0)$, $\dots$, $e_n = (0, 0, \dots, 0, 1)$; $\{e_1, e_2, \dots, e_n\}$ is readily seen to be a basis for F^n and is called the **standard basis** for F^n . $\blacklozenge$

Example A.1.5. In $P_n(F)$ the set $\{1, x, x^2, \dots, x^n\}$ is a basis. We call this basis the **standard basis** for $P_n(F)$. $\blacklozenge$

As we were able to express every point in $\mathbb{R}^n$ as a linear combination of axis uniquely, we can do the same thing within basis:

Theorem A.2. Let V be a vector space and $\beta = \{u_1, u_2, \dots, u_n\}$ be a subset of V . Then β is a basis for V if and only if each $v \in V$ can be uniquely expressed as a linear combination of vectors of β , that is, can be expressed in the form

$$v = a_1 u_1 + a_2 u_2 + \dots + a_n u_n$$

for unique scalars $a_1, a_2, \dots, a_n$.

Proof. Use the definition of basis and linear independence. Choose two representations a and b , and subtract them. Look at the coefficients of $b - a$. $\square$

Also, recall that we can extend the axis in the Euclidean space. Consider the xy -plane and $\mathbb{R}^3$. Note that $\{e_1, e_2\}$ is a basis of xy -plane. It is clear that one can “extend” the axis by choosing any vector which is not in the xy -plane. We can do the same thing in general vector spaces:

Theorem A.3 (Replacement Theorem). Let V be a vector space that is generated by a set G containing exactly n vectors, and let L be a linearly independent subset of V containing exactly m vectors. Then $m \leq n$ and there exists a subset H of G containing exactly $n - m$ vectors such that $L \cup H$ generates V .

Proof. Omit. See the theorem 1.10 of Friedberg Linear Algebra. $\square$

Theorem A.4. Let V be a vector space having a finite basis. Then every basis for V contains the same number of vectors.

Proof. Suppose that β is a finite basis for V that contains exactly n vectors, and let γ be any other basis for V . If γ contains more than n vectors, then we can select a subset S of γ containing exactly $n + 1$ vectors. Since S is linearly independent and β generates V , the replacement theorem implies that $n + 1 \leq n$, a contradiction. Therefore γ is finite, and the number m of vectors in γ satisfies $m \leq n$. Reversing the roles of β and γ and arguing as above, we obtain $n \leq m$. Hence $m = n$. $\square$

This theorem is saying that the number elements of bases are same with respect to each vector spaces. Therefore, we may define the dimension with this:

Definition A.6. A vector space is called **finite-dimensional** if it has a basis consisting of a finite number of vectors. The unique number of vectors in each basis for V is called the **dimension** of V and is denoted by $\dim(V)$. A vector space that is not finite-dimensional is called **infinite-dimensional**.

For later, we define the following:

Definition A.7. Let V and W be vector spaces. We say that V is **isomorphic** to W if there exists a linear transformation $T : V \rightarrow W$ that is invertible. Such a linear transformation is called an **isomorphism** from V onto W .

Example A.1.6. Define $T : F^2 \rightarrow P_1(F)$ by $T(a_1, a_2) = a_1 + a_2x$. It is easily checked that T is an isomorphism; so F^2 is isomorphic to $P_1(F)$. ♦

A.2 Dual Spaces

Let V be a vector space over a field F . In this part, we want to study about *linear transformations* such that:

$$T : V \rightarrow F.$$

One may check that the set $\{T : V \rightarrow F : T \text{ is a linear transformation}\}$ is a vector space over F .

Example A.2.1. Consider $\mathbb{R}^n$, and let $\beta = \{e_1, e_2, \dots, e_n\}$ be the standard ordered basis for $\mathbb{R}^n$. For each $i = 1, 2, \dots, n$, define $f_i(x) = a_i$, where

$$x = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}$$

is the coordinate vector of x relative to β . Then f_i is a linear functional on $\mathbb{R}^n$. ♦

Example A.2.2. More generally, let V be a finite-dimensional vector space, and let $\beta = \{x_1, x_2, \dots, x_n\}$ be an ordered basis for V . For each $i = 1, 2, \dots, n$, define $f_i(x) = a_i$, where x can be represented by:

$$x = a_1x_1 + a_2x_2 + \cdots + a_nx_n$$

with respect to β . Then f_i is a linear functional on V called the ***ith coordinate function with respect to the basis β*** . ♦

Definition A.8. For a vector space V over F , we define the ***dual space*** of V to be the vector space $\{T : V \rightarrow F : T \text{ is a linear transformation}\}$, denoted by V^* .

Fact. $\dim V = \dim V^*$.

Then we can observe that the coordinate functions are indeed basis elements of the dual space.

Theorem A.5. Suppose that V is a finite-dimensional vector space with the ordered basis $\beta = \{x_1, x_2, \dots, x_n\}$. Let f_i ($1 \leq i \leq n$) be the *ith* coordinate function with respect to β as just defined, and let $\beta^* = \{f_1, f_2, \dots, f_n\}$. Then β^* is an ordered basis for V^* , and, for any $f \in V^*$, we have

$$f = \sum_{i=1}^n f(x_i) f_i.$$

Proof. By the fact above, one may show that β^* is a basis. Also, since $f \in V^*$ is linear, we can clearly represent f with β^* as the given form. □